



Pakistan

WUENIC 2023 revision,
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World Health
Organization

WHO/UNICEF Estimates of National Immunization Coverage (WUENIC), 2023 revision

Every year, WHO and UNICEF jointly review submissions from Member States on national immunization coverage, including annual administrative and official coverage, finalized survey reports and data from both published and grey literature. The data is triangulated with consideration of potential biases and local expert opinions to differentiate between accurately reflective empirical data and potentially misleading data, to assess the most likely coverage levels for each country.

WHO and UNICEF produce country-specific estimates by individually reviewing each country's data without borrowing from other countries in the absence of data. These estimates are not based on ad hoc adjustments to reported data; sometimes, empirical data come from a single source, typically nationally reported coverage data. If no data are available for a specific country-vaccine-year combination, data from earlier and later years are considered and interpolated to fill in the gaps. When data sources are conflicting and show significant variations, efforts are made to determine the most likely estimate, taking into account potential biases in the available data.

This slide deck presents the latest WUENIC estimates (published 15 July 2024) for Pakistan.

Methods:

- [Burton et al. 2009. WHO and UNICEF estimates of national infant immunization coverage: methods and processes.](#)
- [Burton et al. 2012. A formal representation of the WHO and UNICEF estimates of national immunization coverage: a computational logic approach.](#)
- [Brown et al. 2013. An introduction to the grade of confidence used to characterize uncertainty around the WHO and UNICEF estimates of national immunization coverage.](#)

Definitions of immunization terms

Vaccine coverage

Percentage of infants (children under one year of age) who received certain vaccine-doses. For example, coverage of DTP3 is the percentage of infants who received all three doses of diphtheria, tetanus, and pertussis (DTP) vaccine.

Unvaccinated

An infant that did not receive the first dose of a vaccine series. The term "zero-dose" is used to describe children unvaccinated with DTP1.

Under-vaccinated

An infant who received some but not all the recommended vaccine-doses in the national schedule.

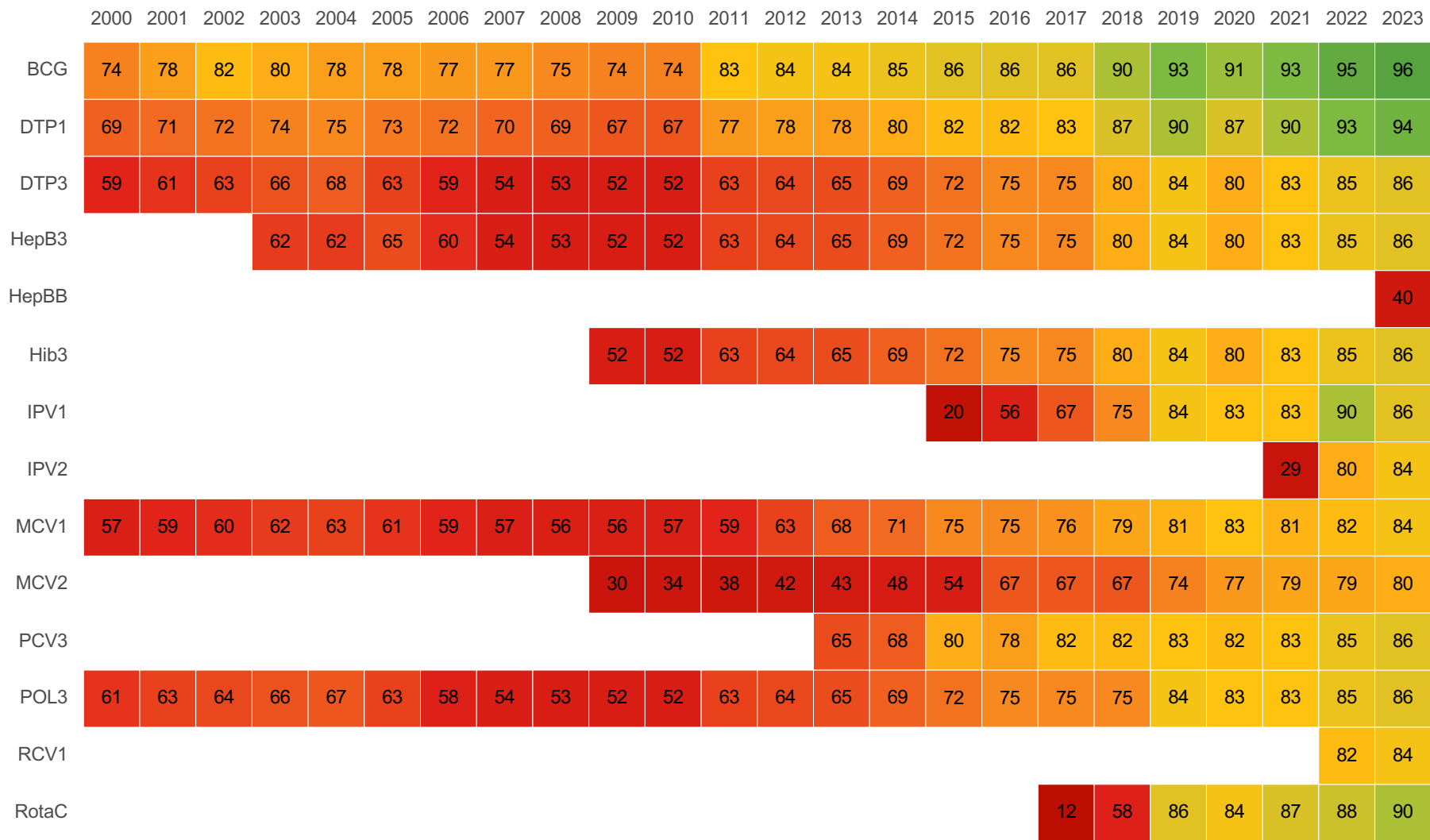
Vaccine-Doses

- Bacillus Calmette-Guerin (BCG): vaccine against tuberculosis
- Hepatitis B birth dose, given within 24 hours after birth (HepBB)
- Diphtheria, tetanus, and pertussis vaccine, first dose (DTP1) and third dose (DTP3)
- Hepatitis B vaccine, third dose (HepB3)
- Haemophilus influenzae type B vaccine, third dose (Hib3)
- Poliomyelitis vaccine, third dose (Polio3)
- Inactivated polio vaccine, first dose (IPV1) and second dose (IPV2): second dose is only shown for oral polio vaccine (OPV) using countries
- Measles containing vaccine, first dose (MCV1) and second dose (MCV2)
- Rotavirus vaccine, last dose (RotaC)
- Pneumococcal vaccine, third dose (PCV3)
- Yellow Fever vaccine (YFV)
- Meningococcal A vaccine (MengA)
- Human Papillomavirus vaccine, last dose (HPVc): vaccine to protect against certain types of human papillomavirus that can lead to cancer or genital warts

The Immunization Agenda 2030 (IA2030)

The IA2030 is a global strategy endorsed by the World Health Assembly aiming to ensure everyone, everywhere, at every age benefits from vaccines for improved health and well-being by 2030. It focuses on increasing vaccine coverage, equity, sustainability and pandemic preparedness while promoting life-course immunization and integrating immunization with other health services.

Vaccine coverage, Pakistan, 2000-2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision

This heatmap shows trends in vaccine coverage since 2000, with green cells indicating coverage of 90% or more.

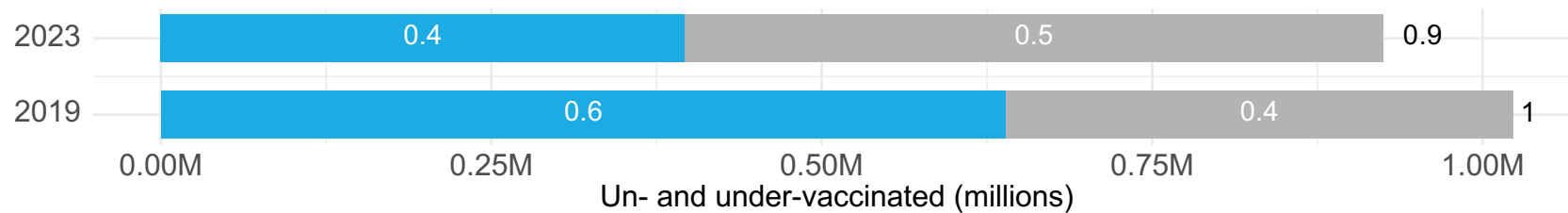
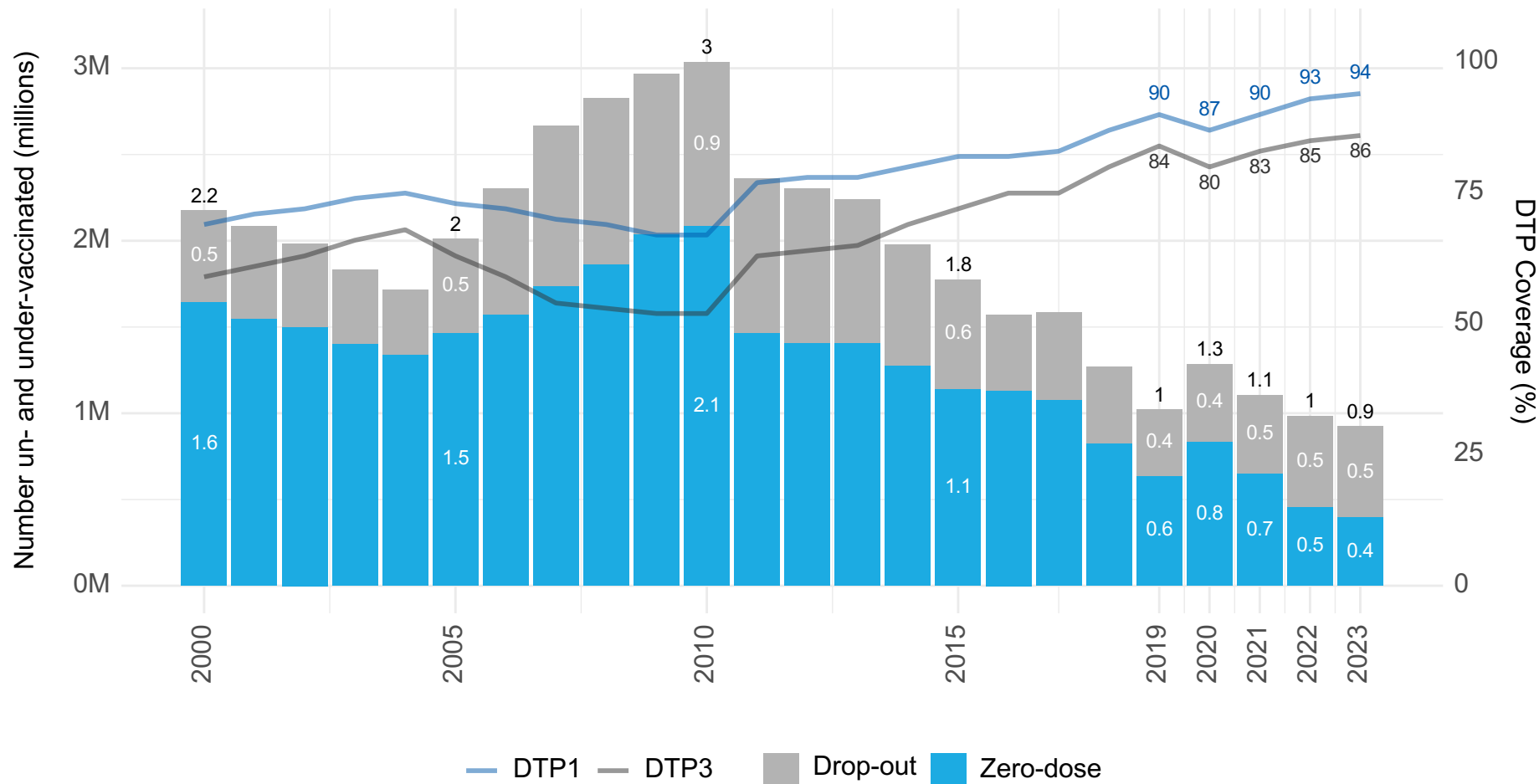
In 2013, none of the 9 vaccines in the national immunization schedule achieved coverage of 90% or more. Vaccine coverage ranged from 43% to 84%.

In 2023, 3 out of 14 (21%) vaccines in the schedule achieved coverage of 90% or more. Vaccine coverage ranged from 40% to 96%.

Since 2003, estimates have been made for 9 new vaccines. HepBB is the newest vaccine reported (2023), which achieved 40% coverage in 2023.

DTP1

Estimated coverage and number of un- and under-vaccinated children for DTP, Pakistan, 2000-2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision

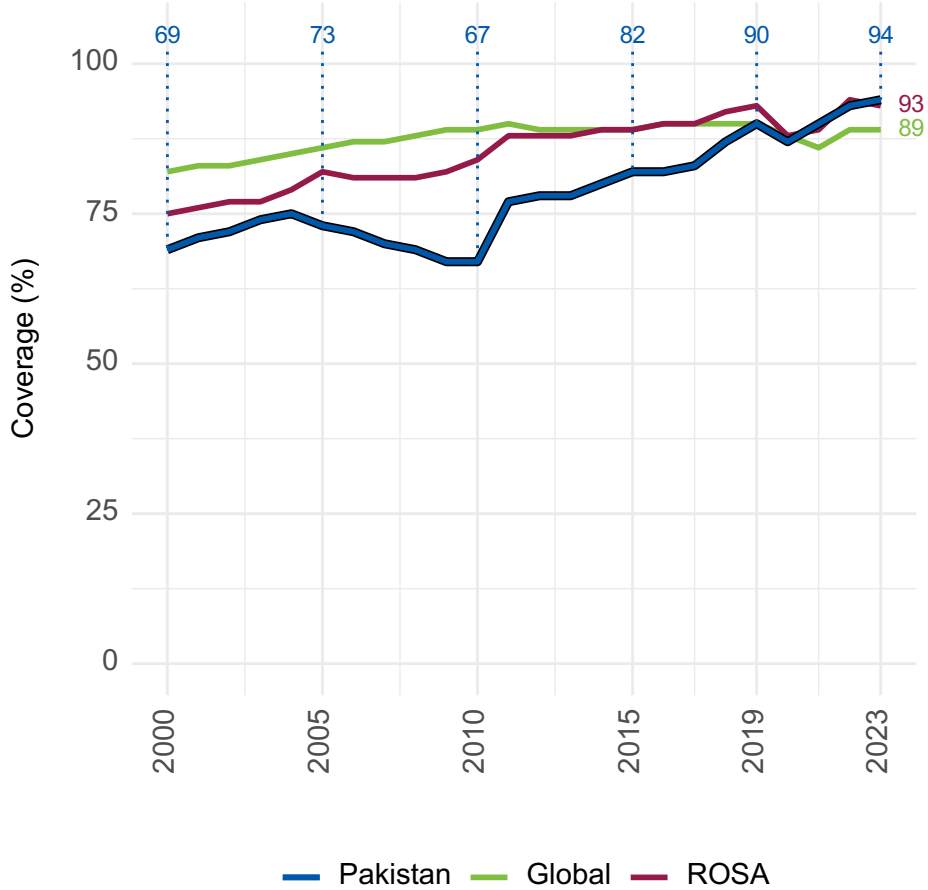
The key goal of the Immunization Agenda 2030 is to make vaccination available to everyone, everywhere, by 2030.

This chart shows diphtheria, tetanus and pertussis-containing vaccine first (DTP1) and third dose (DTP3) coverage trends, the number of zero-dose children and DTP drop-out in Pakistan.

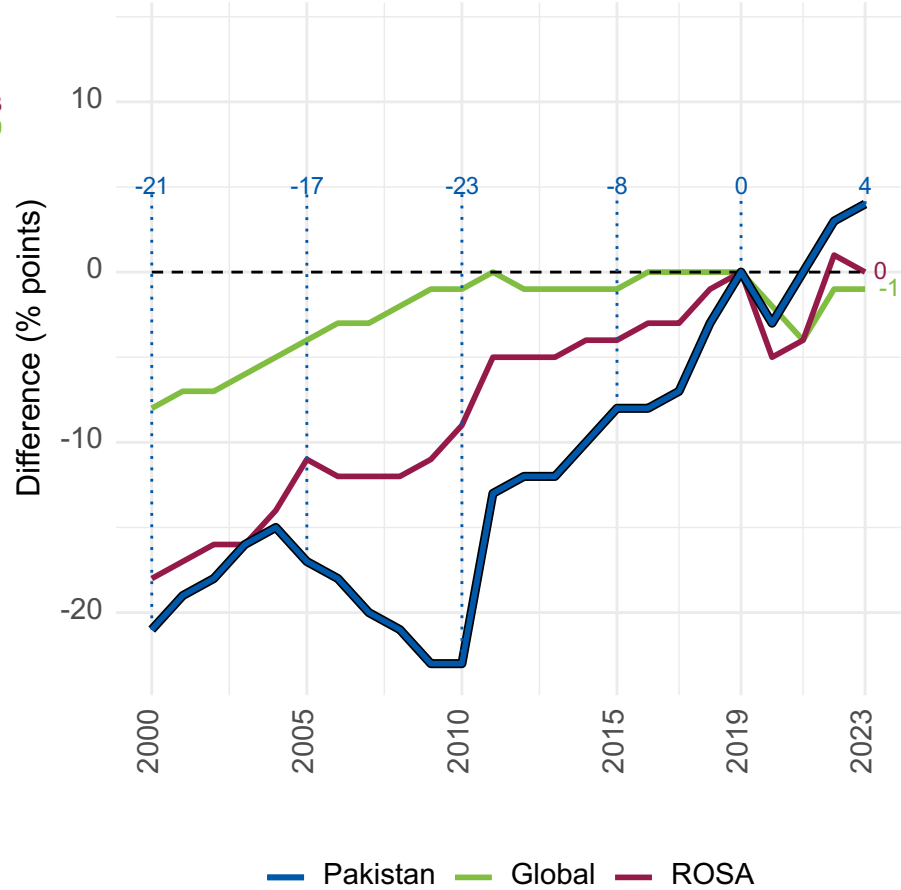
In 2023, DTP1 coverage in Pakistan remained relatively constant within 1% at 94%. The number of children missing out on any DTP vaccination (zero-dose children) improved from 460,000 in 2022 to 396,000 in 2023.

DTP3 coverage remained relatively constant within 1% at 86% in 2023, leaving 925,000 children vulnerable to vaccine-preventable diseases.

DTP1 coverage, Pakistan, 2000-2023



DTP1 coverage difference compared to 2019, Pakistan

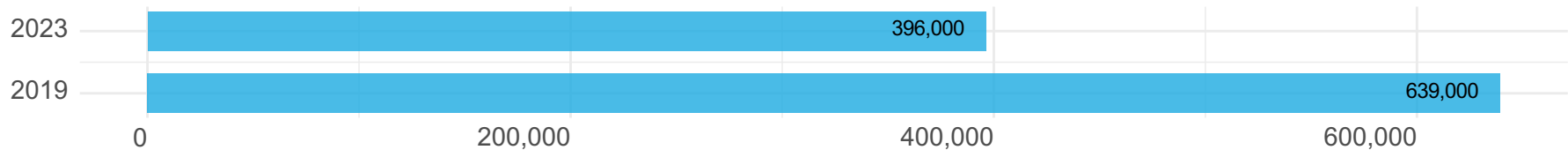


In 2023, DTP1 coverage in Pakistan (94%) was 5 percentage points higher than the global average (89%) and 1 percentage point higher than average across all ROSA countries (93%).

National DTP1 coverage was 4 percentage points higher than in 2019 (90%).

This equates to 396,000 zero-dose children in 2023 compared to 639,000 zero-dose children in 2019.

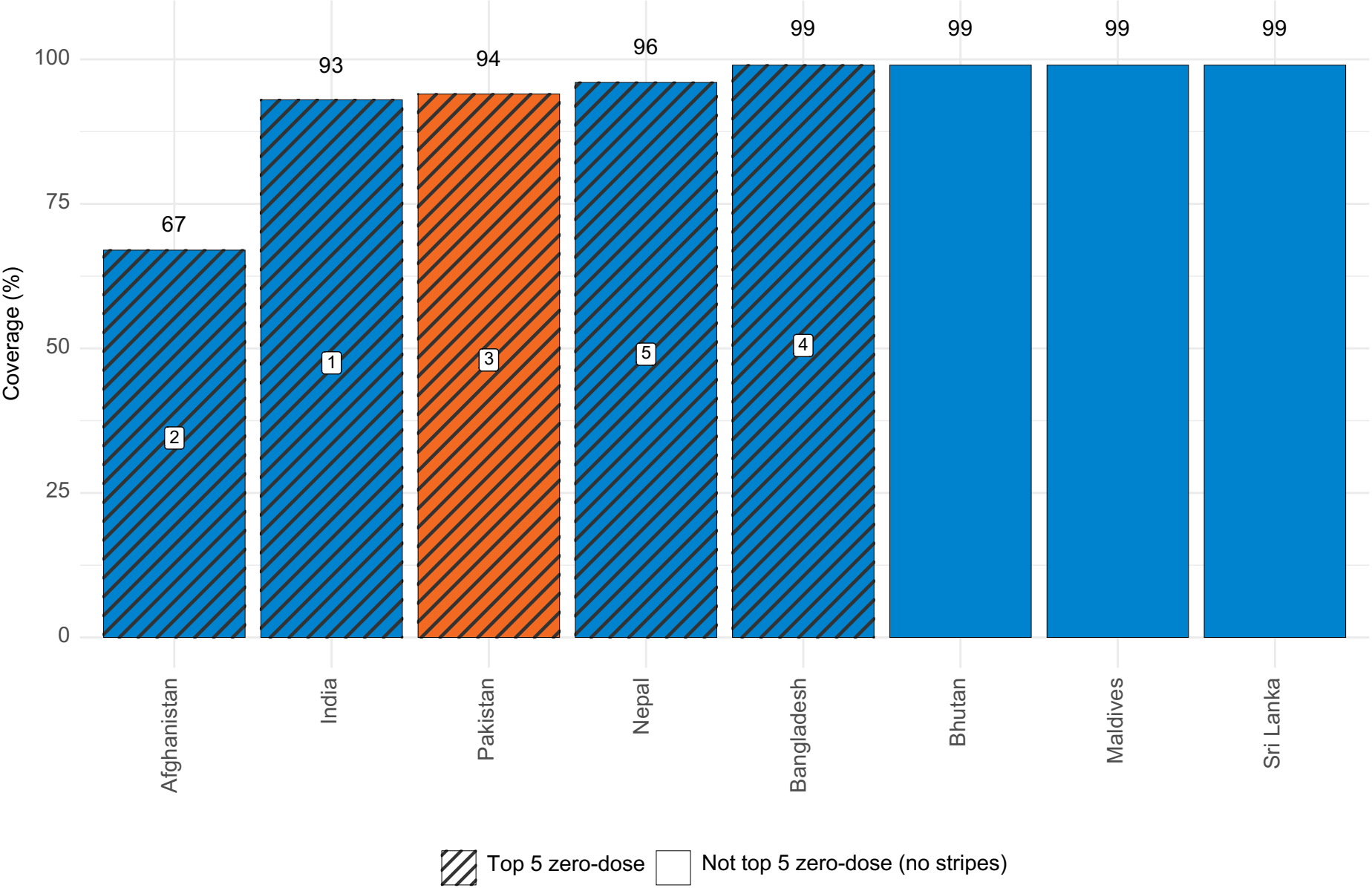
Number of zero-dose children, Pakistan, 2019 and 2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision

Note: Coverage difference compared to 2019 - values above zero indicate coverage higher than in 2019 and values below zero indicate coverage lower than in 2019.

DTP1 coverage and ranking of number zero-dose, by country, ROSA, 2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision
Note: Bars are ranked by ascending coverage. Numbers in bubbles display top 5 rank based on absolute number of zero-dose children.

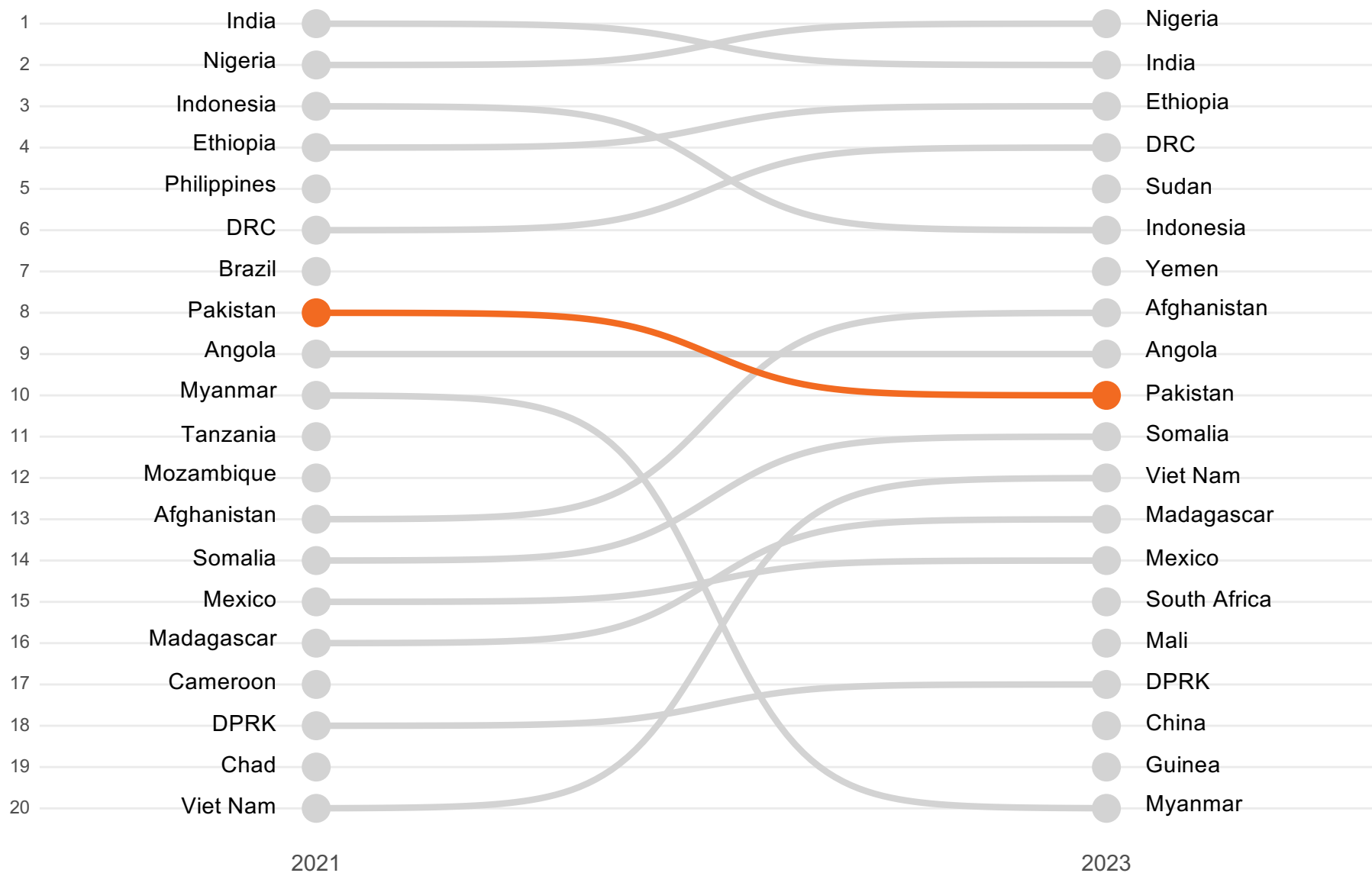
This chart shows DTP1 coverage in countries in ROSA from lowest to highest coverage, and the rank of the top 5 countries with the most zero-dose children, based on absolute numbers.

In 2023, Pakistan ranked number 3 out of 8 countries for lowest DTP1 coverage (based on tied ranks).

Pakistan was in the top 5 countries with the most zero-dose children (rank=3).

Note: Large cohort countries may have high numbers of zero-dose children despite high vaccine coverage. It is important to consider both coverage and absolute numbers of unvaccinated children to ensure vulnerable countries with small birth cohorts are not overlooked.

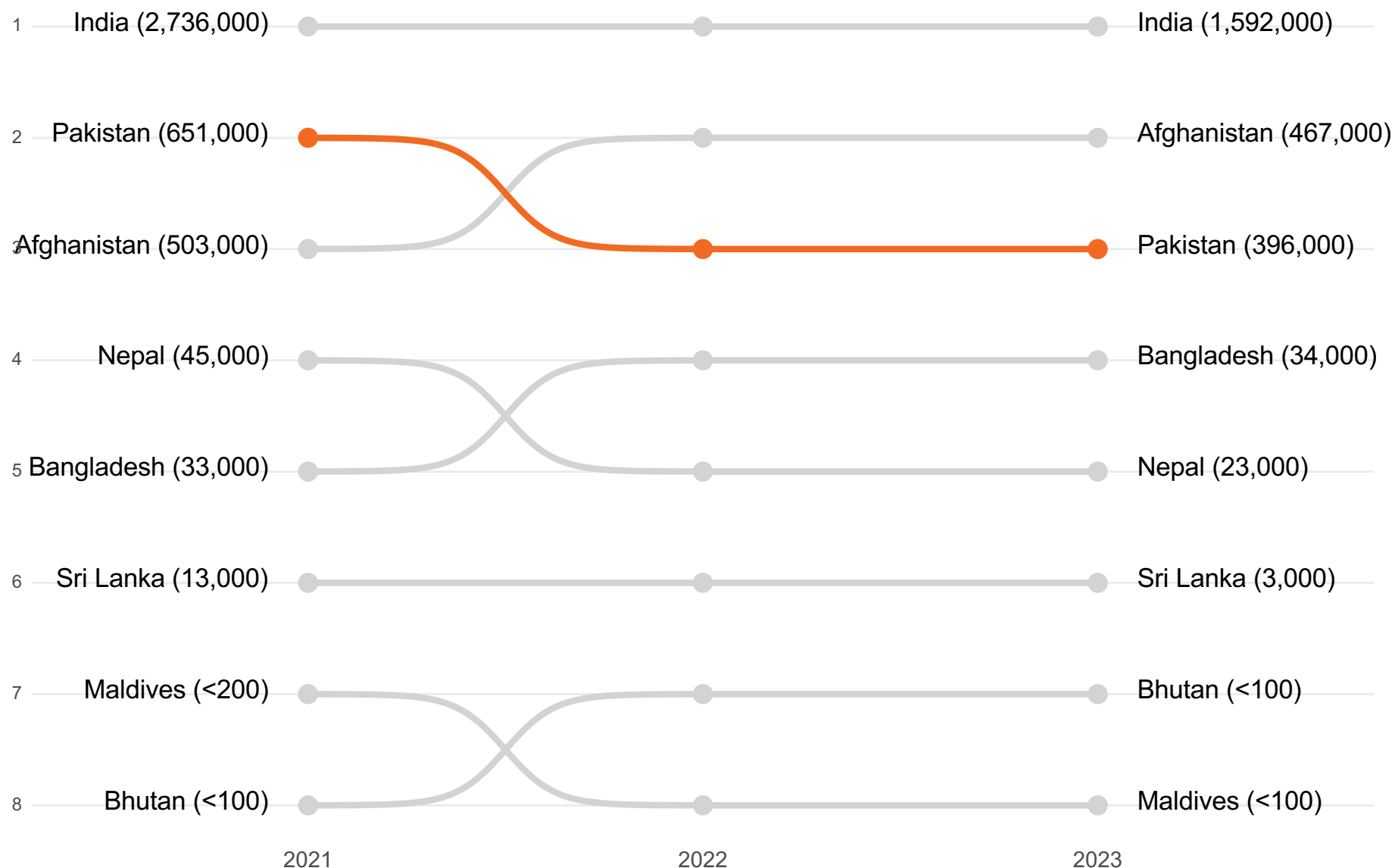
IA2030 top 20 zero-dose countries rank, 2021-2023



20 countries were prioritised in the context of the Immunization Agenda 2030 (IA2030), based on their number of zero-dose children in 2021. This chart compares the ranking of the top 20 countries in 2021 and 2023. Rank 1 indicates the country with the highest absolute number of zero-dose children.

Pakistan was in the top 20 countries in 2021 and 2023.

Countries ranked by number of zero-dose children, ROSA, 2021-2023



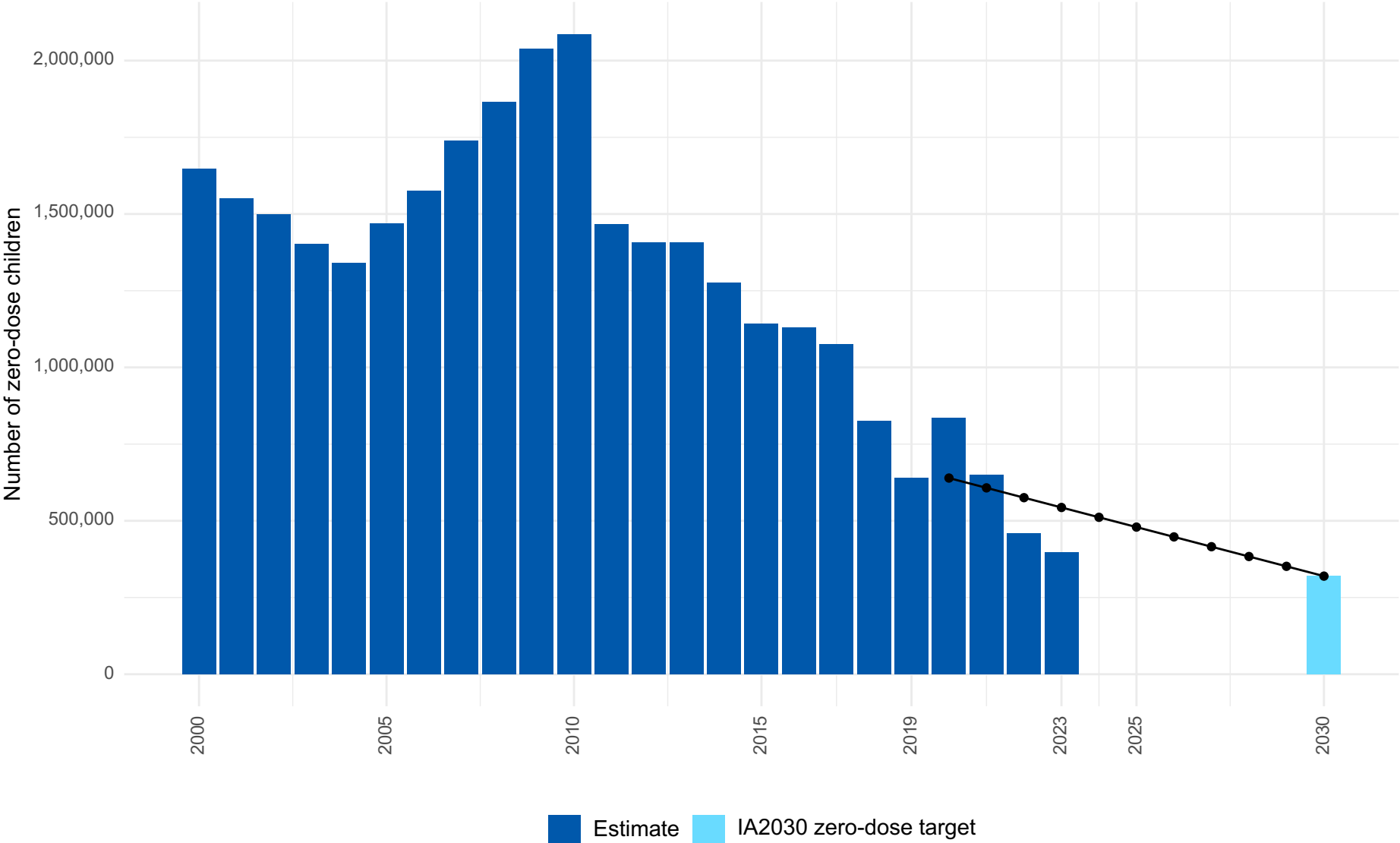
This chart compares the ranking of countries in ROSA based on the absolute number of zero-dose children, with rank 1 representing the country with the most zero-dose children.

In 2021, Pakistan ranked number 2 out of 8 countries with 651,000 zero-dose children.

In 2023, Pakistan ranked number 3 out of 8 countries with 396,000 zero-dose children.

Note: Absolute numbers of zero-dose children is based on a combination of programme performance and surviving infant target population size. Countries may climb to a higher rank despite a decline in number of zero-dose children as the ranking also depends on performance of other countries in the region.

Estimated number of zero-dose children, 2000-2023 and target by 2030, Pakistan



IA2030 aims to leave no one behind with immunization and calls on all countries to reduce the number of zero dose children by half by 2030.

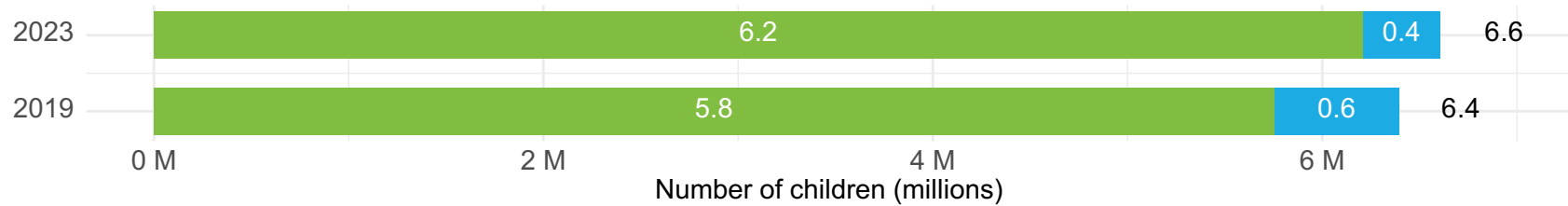
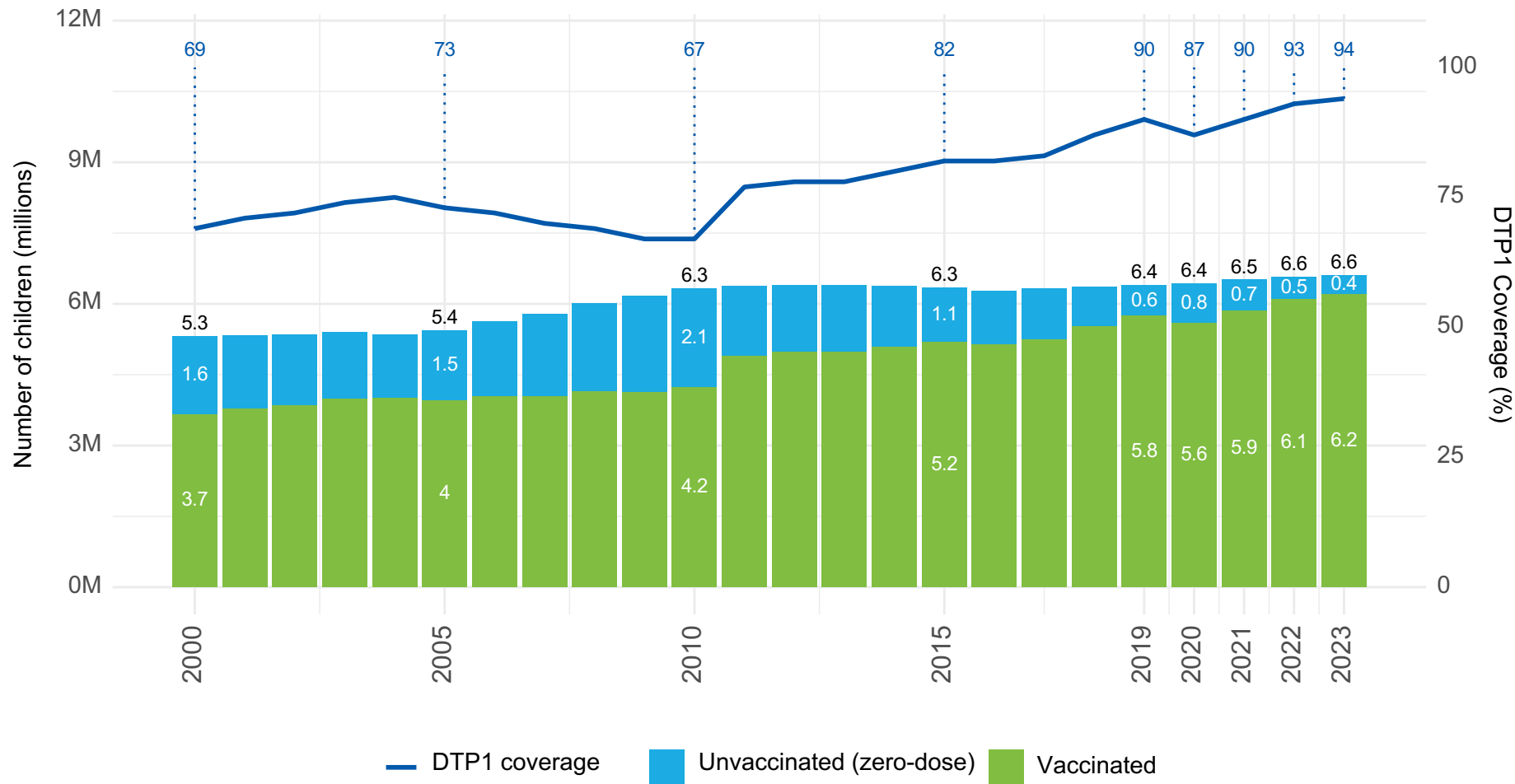
- This chart shows:
- Estimated number of zero-dose children in 2000-2023 (dark blue bars)
 - Zero-dose target by 2030 (light blue bar)
 - Annual goals to reach the 2030 target based on a linear trajectory of decline (points)

In 2023, the number of zero-dose children was approximately 27% lower than (ie. the country had achieved) the annual goal.

Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision.

Note: The Immunization Agenda 2030 (IA2030) calls on all countries to reduce the number of zero dose children in 2019 by half by 2030. Dark blue bars are the estimated number of zero-dose children in 2000-2023, light blue bar is the target number of zero-dose children by 2030. Line shows trajectory the country needs to be on, and points show annual goals to meet the target by 2030, assuming a linear decline.

Estimated DTP1 coverage, and number of vaccinated and unvaccinated children, Pakistan, 2000-2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision

DTP1 coverage in 2023 (94%) was higher than in 2019 (90%).

The number of children vaccinated with DTP1 increased 8% compared to in 2019.

The number of surviving infants increased approximately 3% compared to in 2019.

In 2023, 500,000 more children were vaccinated than in 2019.

In 2023, there were 200,000 more surviving infants (target population) than in 2019.

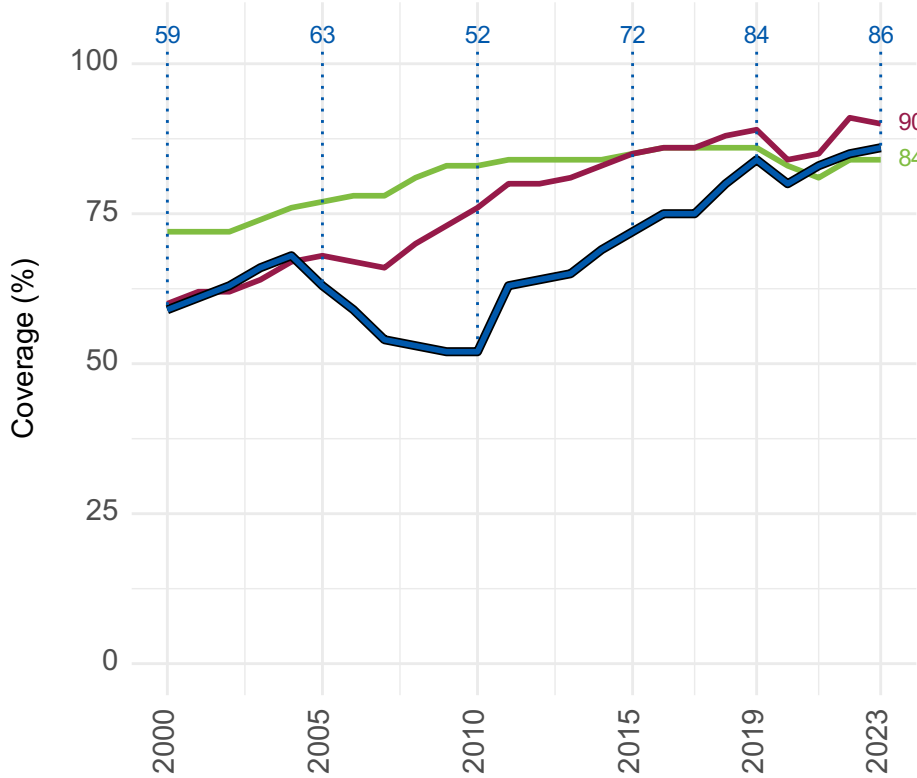
For vaccine coverage to increase, the number of children vaccinated must increase at a faster rate than the population increases.



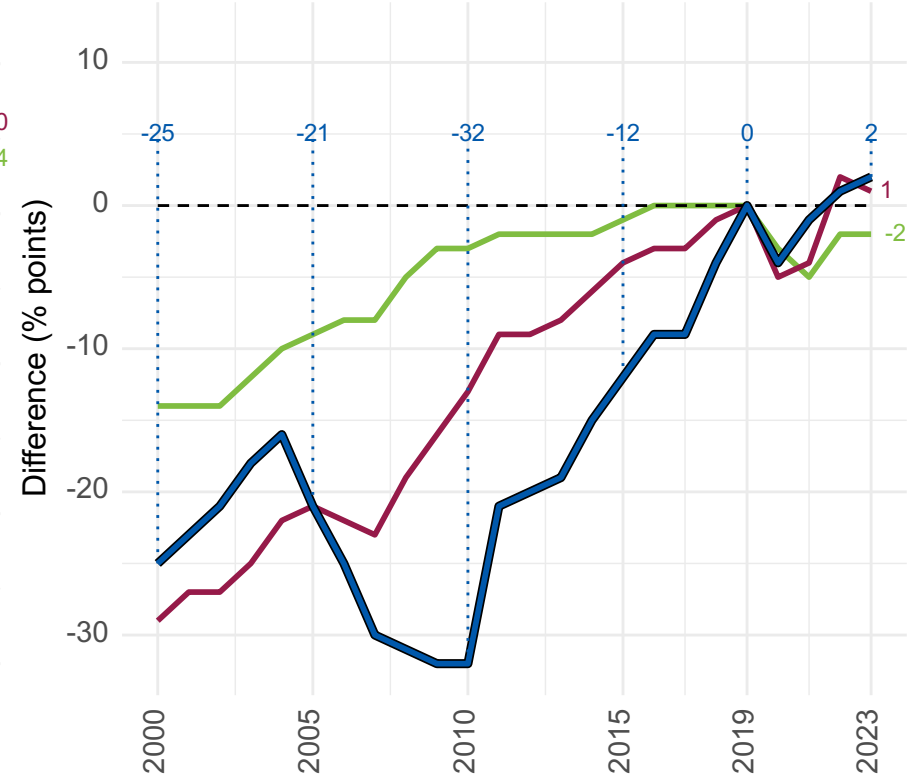
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DTP3

DTP3 coverage, Pakistan, 2000-2023



DTP3 coverage difference compared to 2019, Pakistan

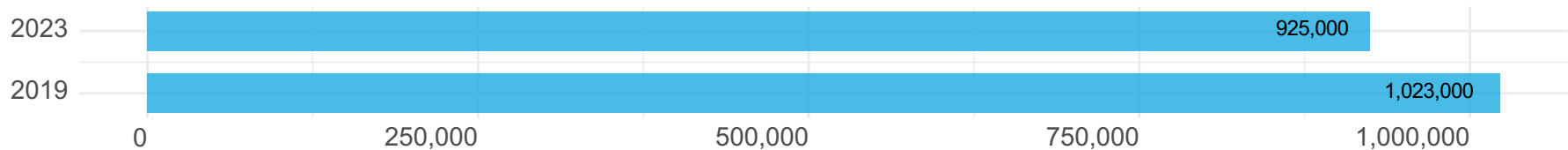


In 2023, DTP3 coverage in Pakistan (86%) was 2 percentage points higher than the global average (84%) and 4 percentage points lower than the average across all ROSA countries (90%).

National DTP3 coverage was 2 percentage points higher than in 2019 (84%).

This equates to 925,000 un- and undervaccinated children in 2023 compared to 1,023,000 un- and undervaccinated children in 2019.

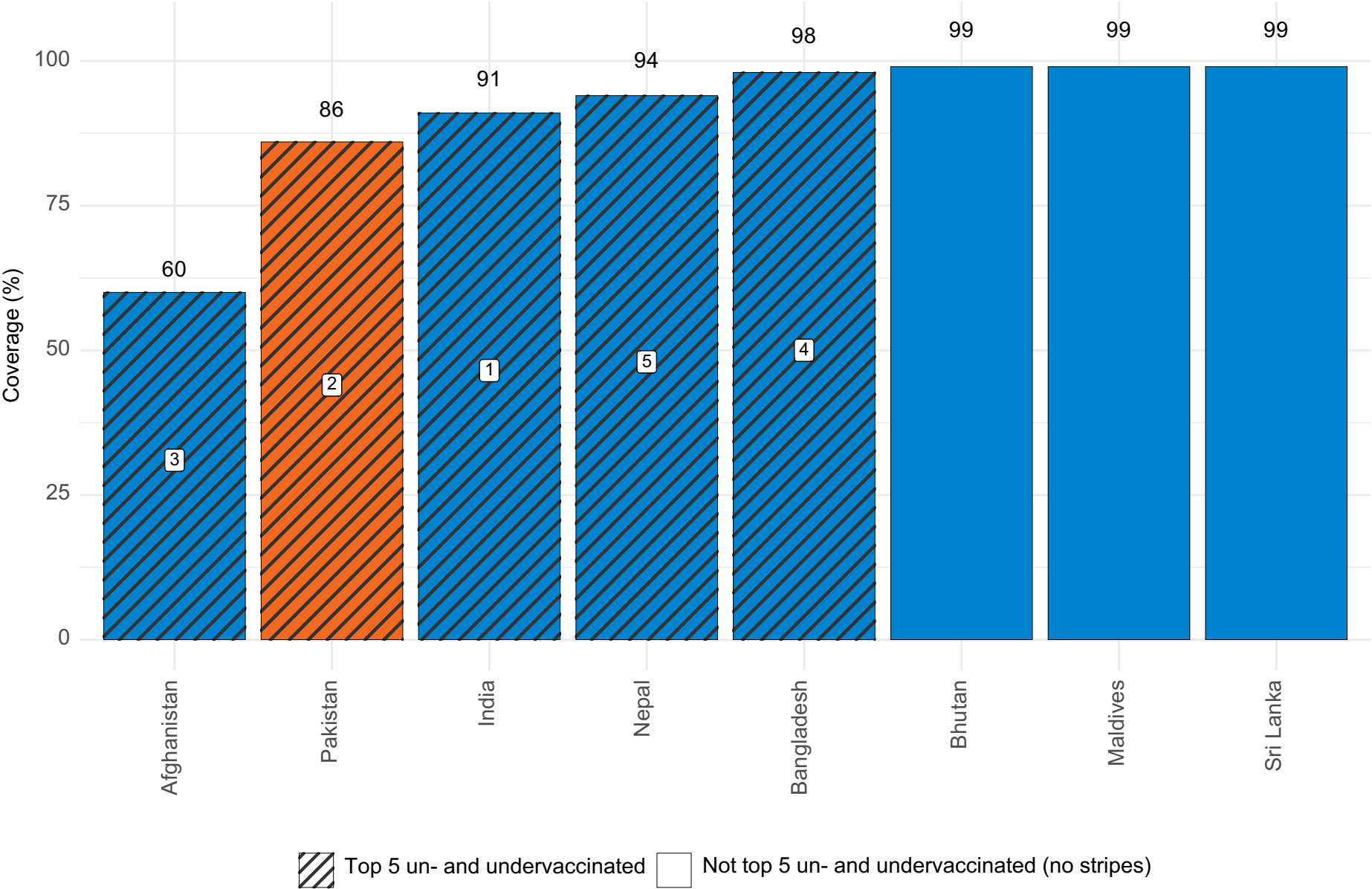
Number of un- and undervaccinated children , Pakistan, 2019 and 2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision

Note: Coverage difference compared to 2019 - values above zero indicate coverage higher than in 2019 and values below zero indicate coverage lower than in 2019.

DTP3 coverage and ranking of number un- and undervaccinated, by country, ROSA, 2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision
Note: Bars are ranked by ascending coverage. Numbers in bubbles display top 5 rank based on absolute number of un- and undervaccinated children.

This chart shows DTP3 coverage in countries in ROSA from lowest to highest coverage, and the rank of the top 5 countries with the most un- and undervaccinated children, based on absolute numbers.

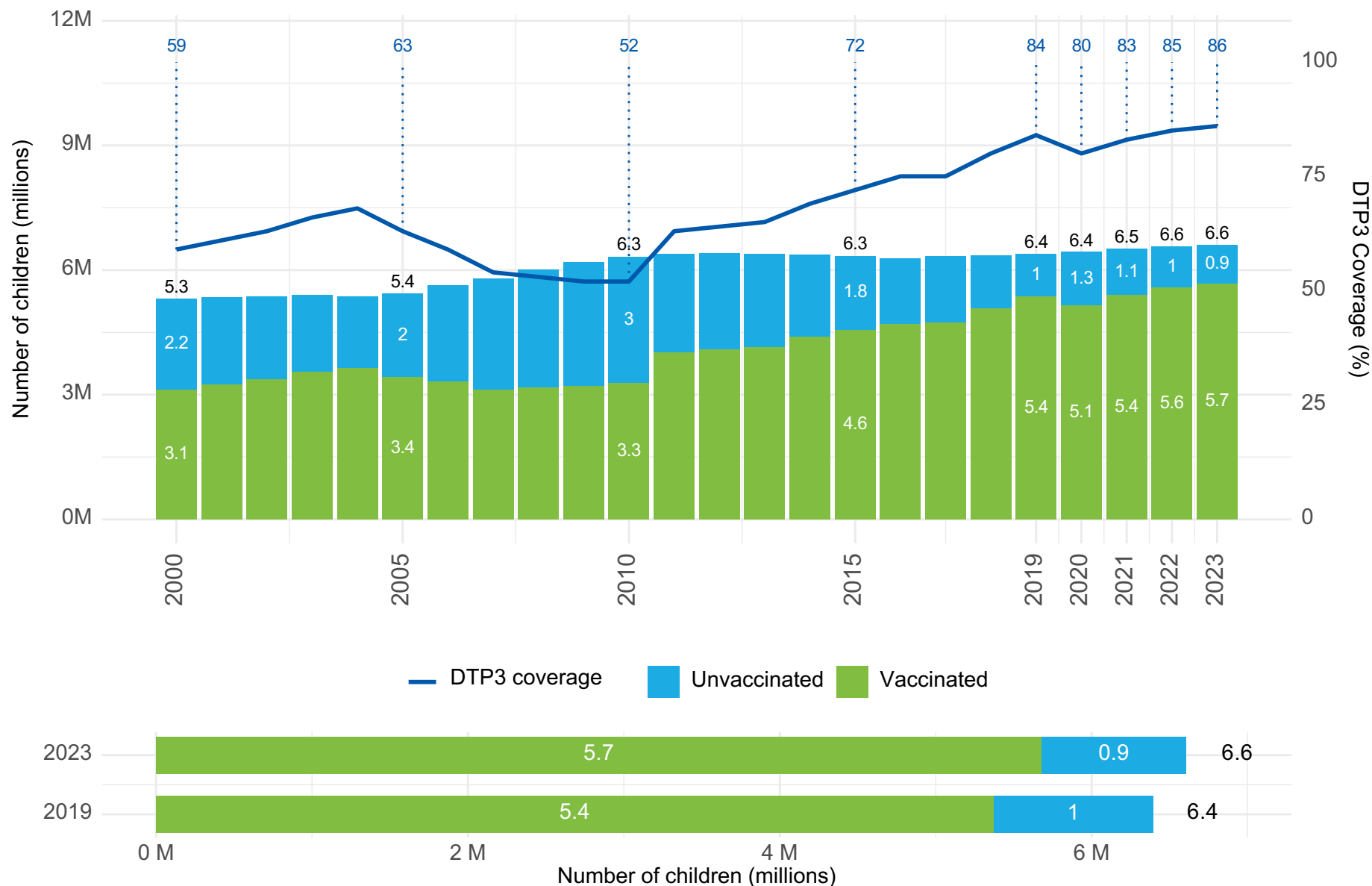
In 2023, Pakistan ranked number 2 out of 8 countries for lowest DTP3 coverage (based on tied ranks).

Pakistan was in the top 5 countries with the most un- and undervaccinated children (rank=2).

Note: Large cohort countries may have high numbers of un- and undervaccinated children despite high vaccine coverage. It is important to consider both coverage and absolute numbers of unvaccinated children to ensure vulnerable countries with small birth cohorts are not overlooked.



Estimated DTP3 coverage, and number of vaccinated and unvaccinated children, Pakistan, 2000-2023



DTP3 coverage in 2023 (86%) was higher than in 2019 (84%).

The number of children vaccinated with DTP3 increased 6% compared to in 2019.

The number of surviving infants increased approximately 3% compared to in 2019.

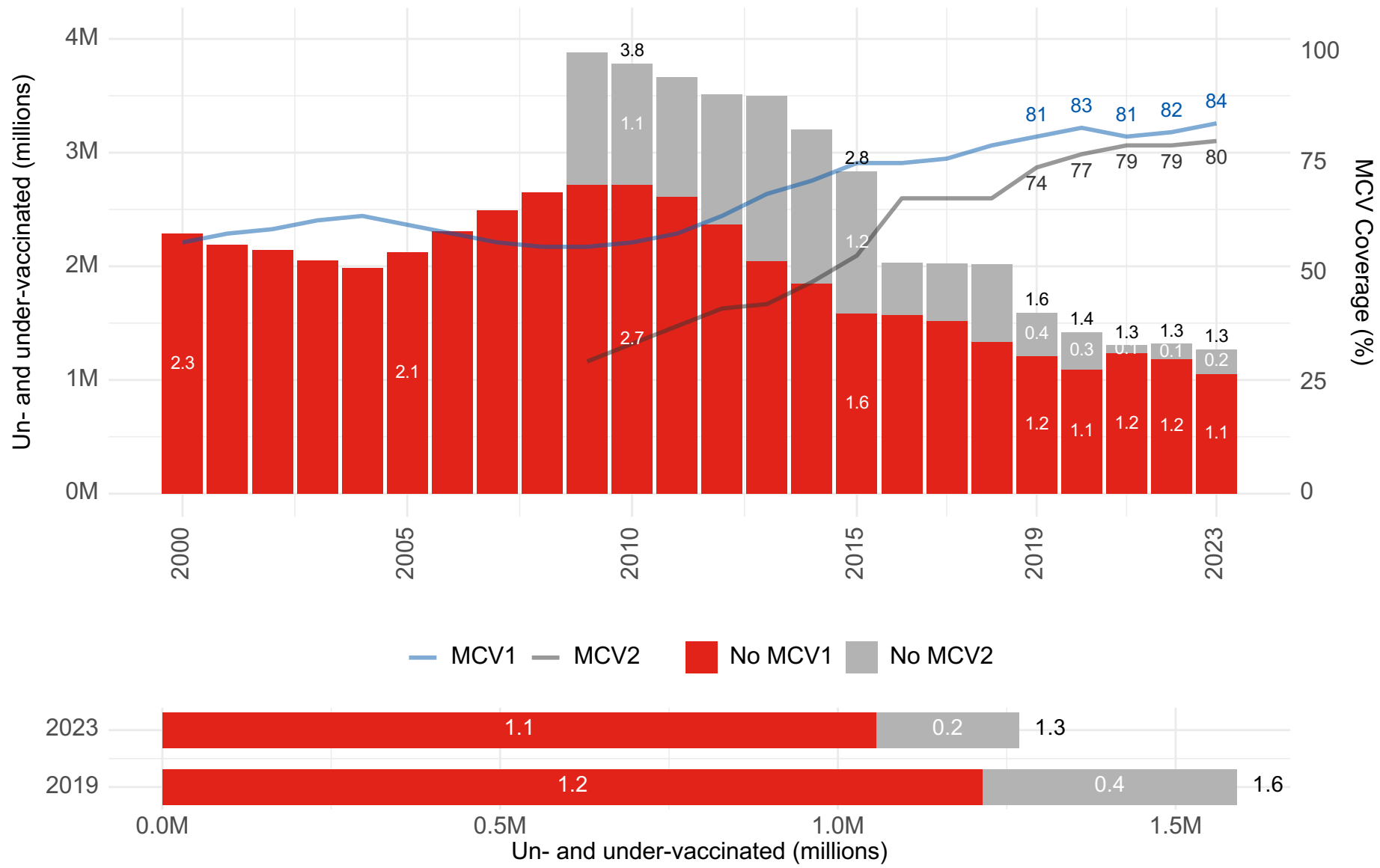
In 2023, 300,000 more children were vaccinated than in 2019.

In 2023, there were 200,000 more surviving infants (target population) than in 2019.

For vaccine coverage to increase, the number of children vaccinated must increase at a faster rate than the population increases.

MCV1

Estimated coverage and number of un- and under-vaccinated children for MCV, Pakistan, 2000-2023



Measles, because of its high transmissibility, acts as a 'canary in the coalmine', quickly exposing any immunity gaps in the population. The coverage of measles containing vaccine (MCV) is thus often used as a tracer for protection.

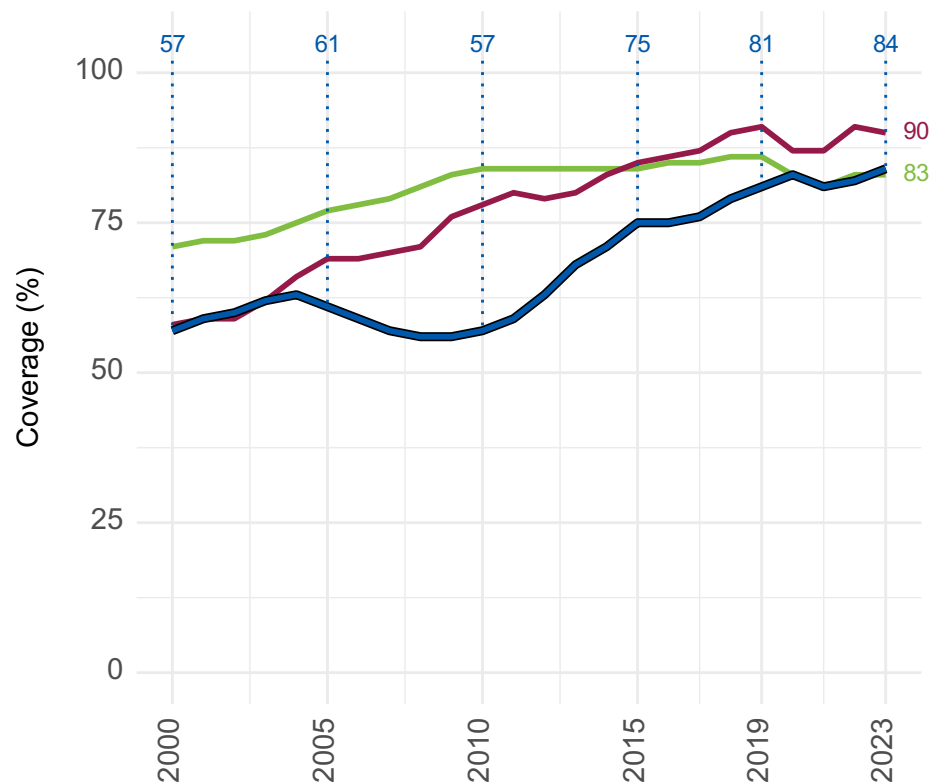
The percentage of children receiving MCV1 – typically at 9 or 12 months depending on the national vaccination schedule – increased to 84%. This is greater than in 2019, where coverage was 81%.

1,057,000 children missed their routine first dose of measles vaccine.

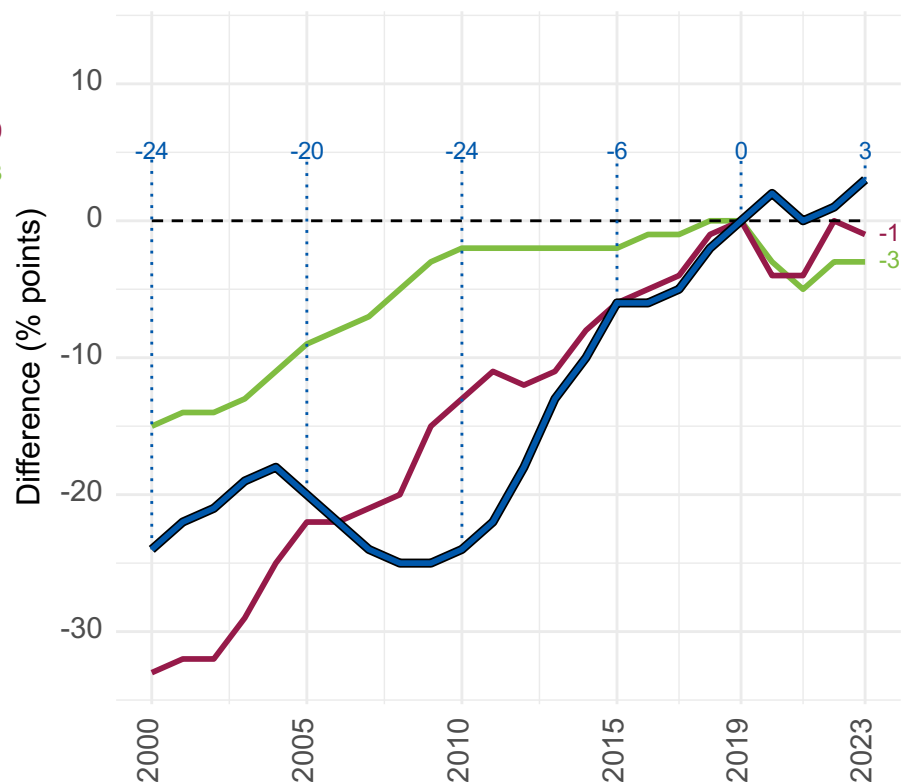
MCV2 is typically administered to children between 18 months and five years old. MCV2 coverage remained relatively constant (within {cvg_noise}%) at 80% in 2023.

Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision
Note: Lines show vaccine coverage and bars show number of children.
There may be discrepancies in total numbers due to rounding.

MCV1 coverage, Pakistan, 2000-2023



MCV1 coverage difference compared to 2019, Pakistan



In 2023, MCV1 coverage in Pakistan (84%) was 1 percentage point higher than the global average (83%) and 6 percentage points lower than the average across all ROSA countries (90%).

National MCV1 coverage was 3 percentage points higher than in 2019 (81%).

This equates to 1,057,000 unvaccinated children in 2023 compared to 1,215,000 unvaccinated children in 2019.

— Pakistan — Global — ROSA

— Pakistan — Global — ROSA

Number of infants unprotected against measles, Pakistan, 2019 and 2023



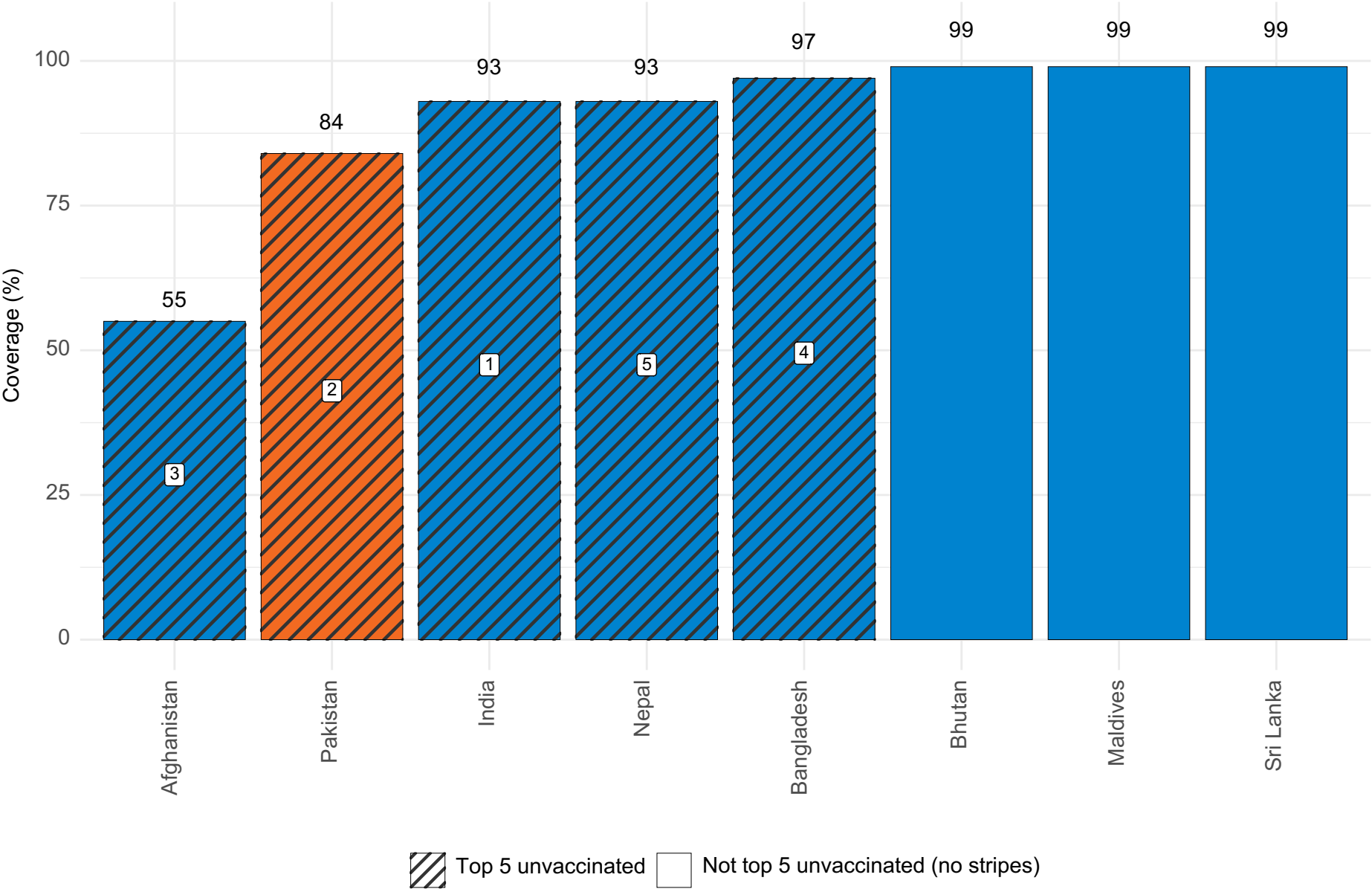
Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision

Note: Coverage difference compared to 2019 - values above zero indicate coverage higher than in 2019 and values below zero indicate coverage lower than in 2019.



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MCV1 coverage and ranking of number unvaccinated, by country, ROSA, 2023



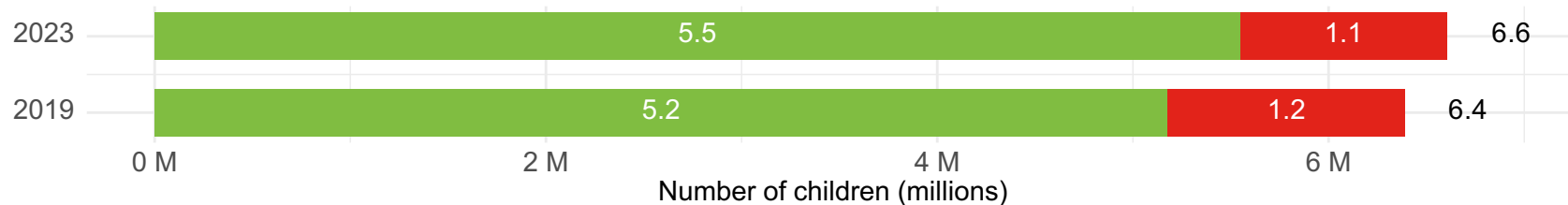
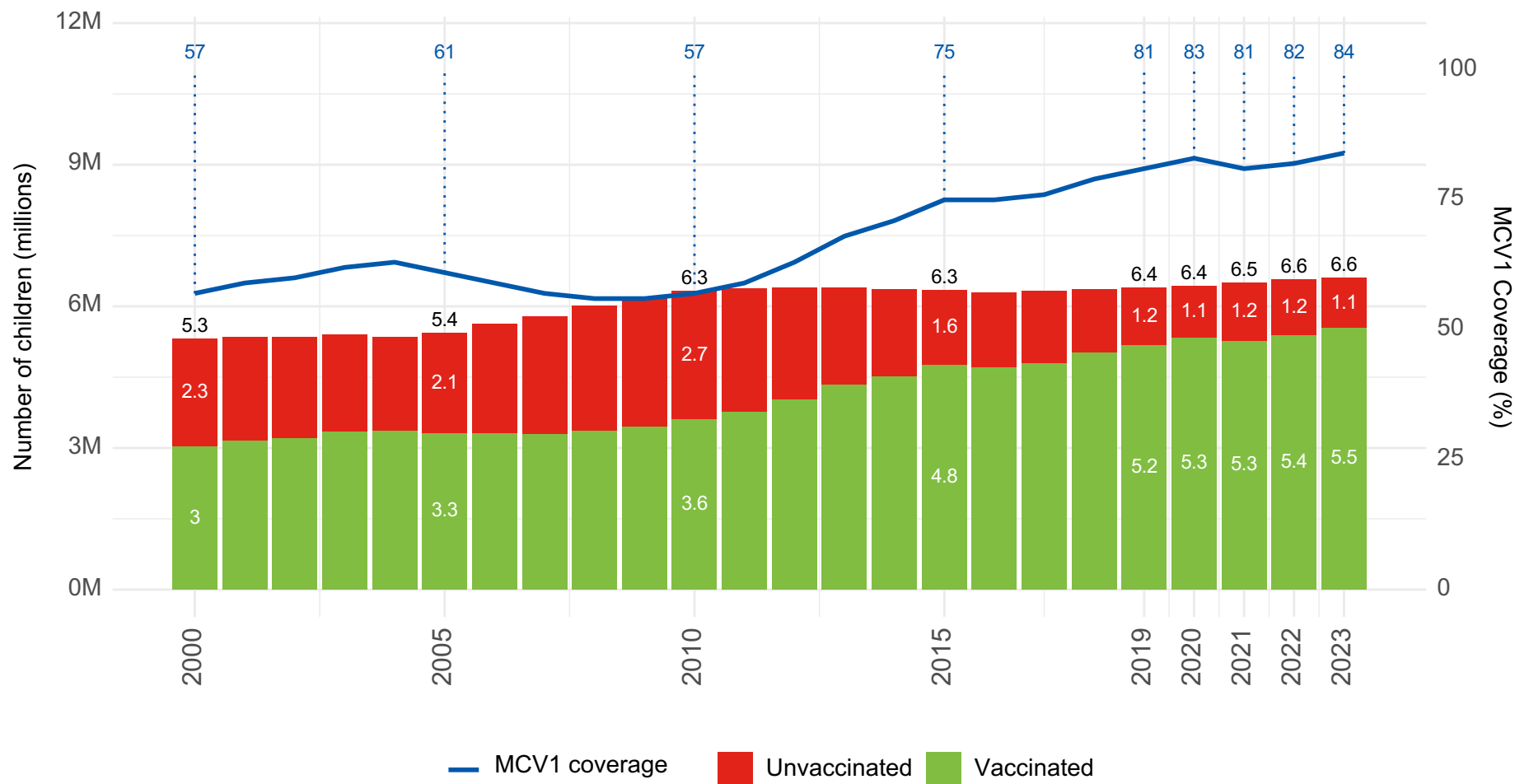
This chart shows MCV1 coverage in countries in ROSA from lowest to highest coverage, and the rank of the top 5 countries with the most unvaccinated children, based on absolute numbers.

In 2023, Pakistan ranked number 2 out of 8 countries for lowest MCV1 coverage (based on tied ranks).

Pakistan was in the top 5 countries with the most unvaccinated children (rank=2).

Note: Large cohort countries may have high numbers of unvaccinated children despite high vaccine coverage. It is important to consider both coverage and absolute numbers of unvaccinated children to ensure vulnerable countries with small birth cohorts are not overlooked.

Estimated MCV1 coverage, and number of vaccinated and unvaccinated children, Pakistan, 2000-2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision

MCV1 coverage in 2023 (84%) was higher than in 2019 (81%).

The number of children vaccinated with MCV1 increased 7% compared to in 2019.

The number of surviving infants increased approximately 3% compared to in 2019.

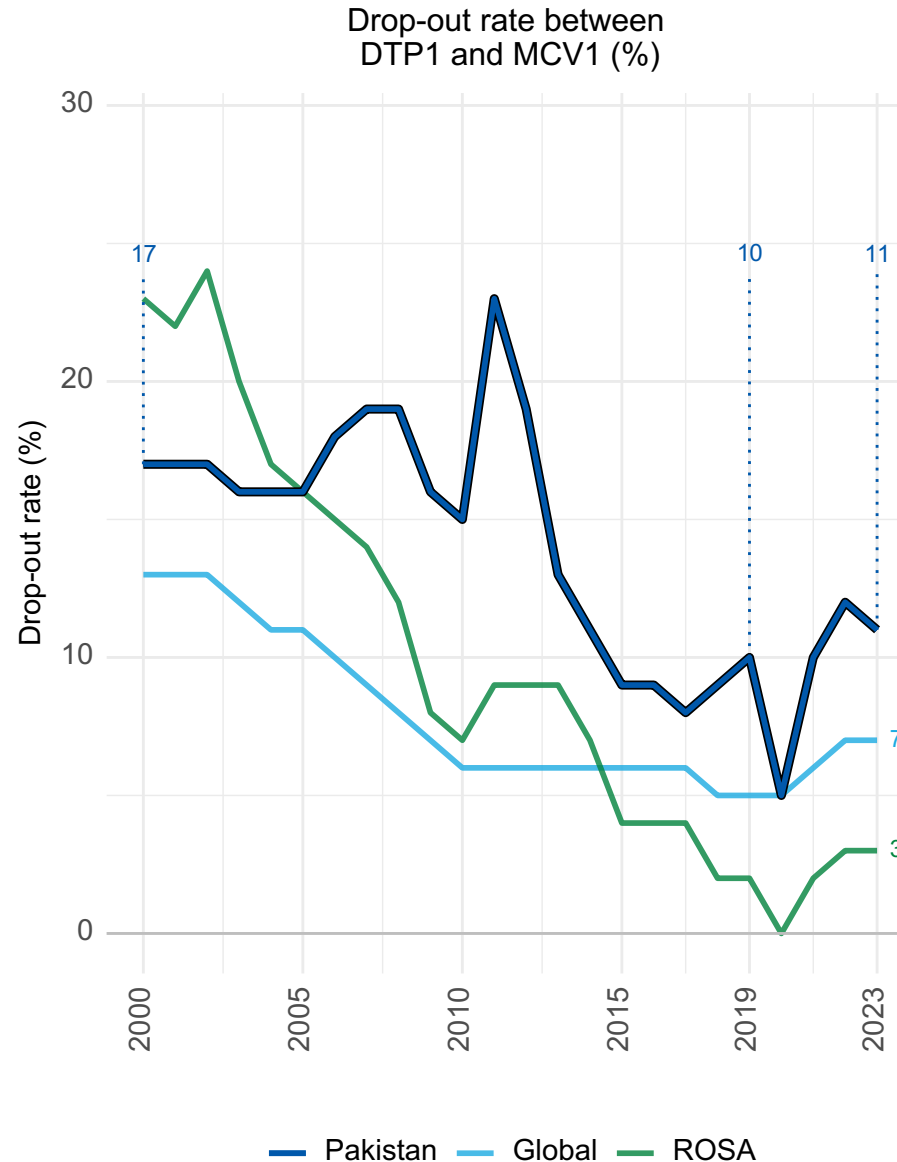
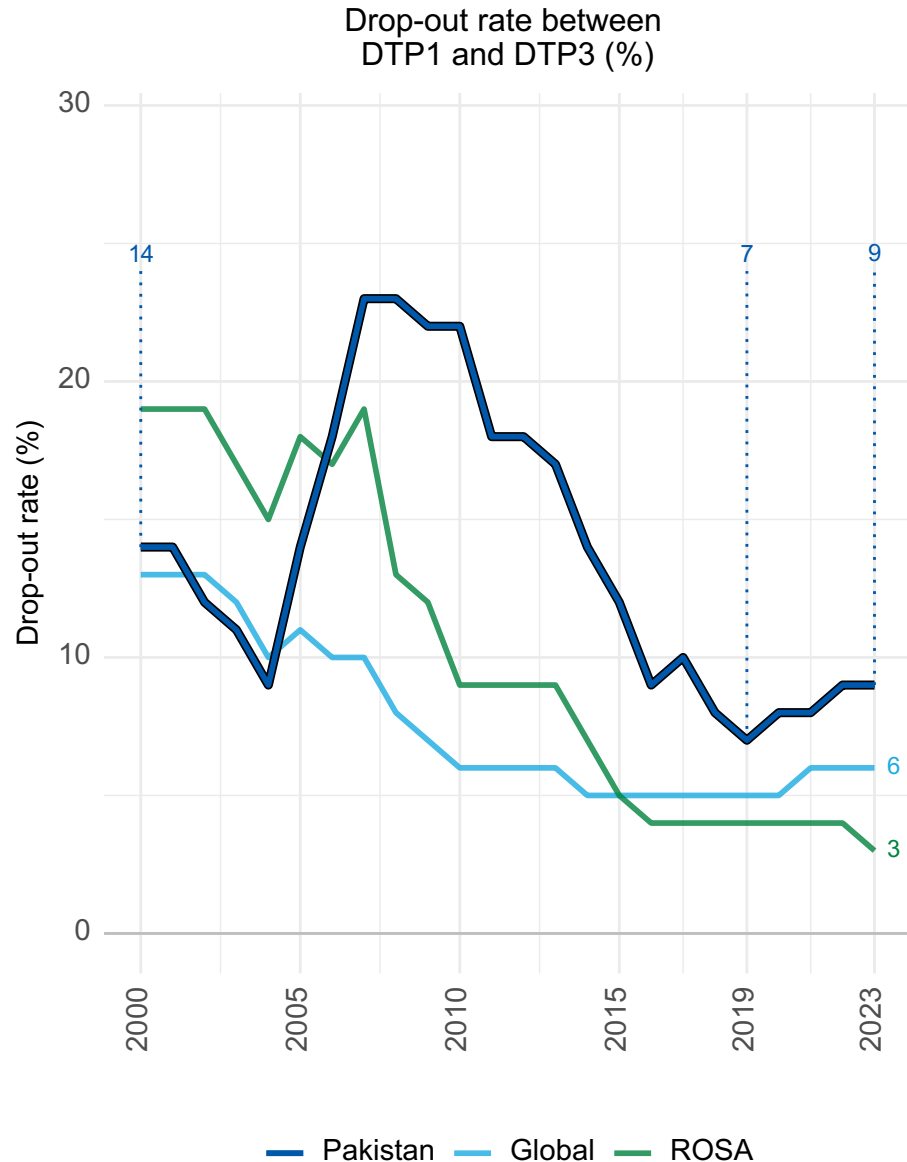
In 2023, 400,000 more children were vaccinated than in 2019.

In 2023, there were 200,000 more surviving infants (target population) than in 2019.

For vaccine coverage to increase, the number of children vaccinated must increase at a faster rate than the population increases.

Childhood immunization: Additional charts

Vaccine drop-out, Pakistan, 2000-2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision
Drop-out classification: <5% = low, 5-10% = medium, >10% = high

Drop-out rates show the percentage of children who received DTP1, but not DTP3/MCV1. Low drop-out rates indicate high retention of children in immunization programmes.

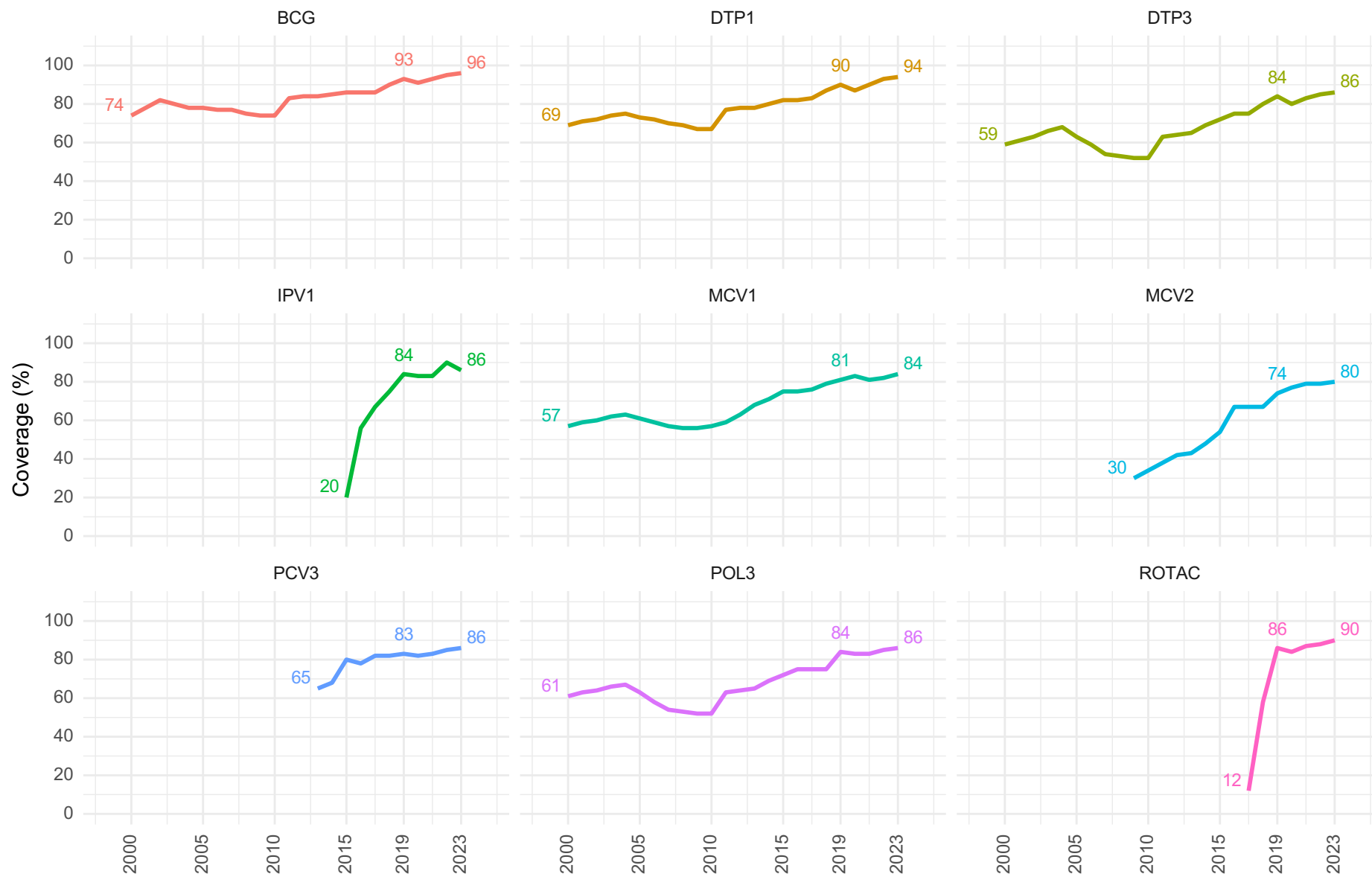
This chart shows trends in drop-out rates between DTP1 and DTP3, and DTP1 and MCV1.

In 2023, 9% of children who received DTP1 did not receive DTP3 (left), and 11% of children who received DTP1 did not receive MCV1 (right).

The medium DTP drop-out rates imply moderate ability to provide a complete series of vaccines early in life. The high DTP-MCV drop-out rates imply poor retention in immunization programmes and ability to provide a full course of vaccines in infancy (up to one year).

In 2023, Pakistan DTP drop-out was higher and DTP-MCV drop-out was higher than global drop-out rates, respectively.

Coverage of recommended childhood vaccines, Pakistan, 2000-2023.



This chart shows trends in coverage of 9 core routine vaccines recommended in childhood.

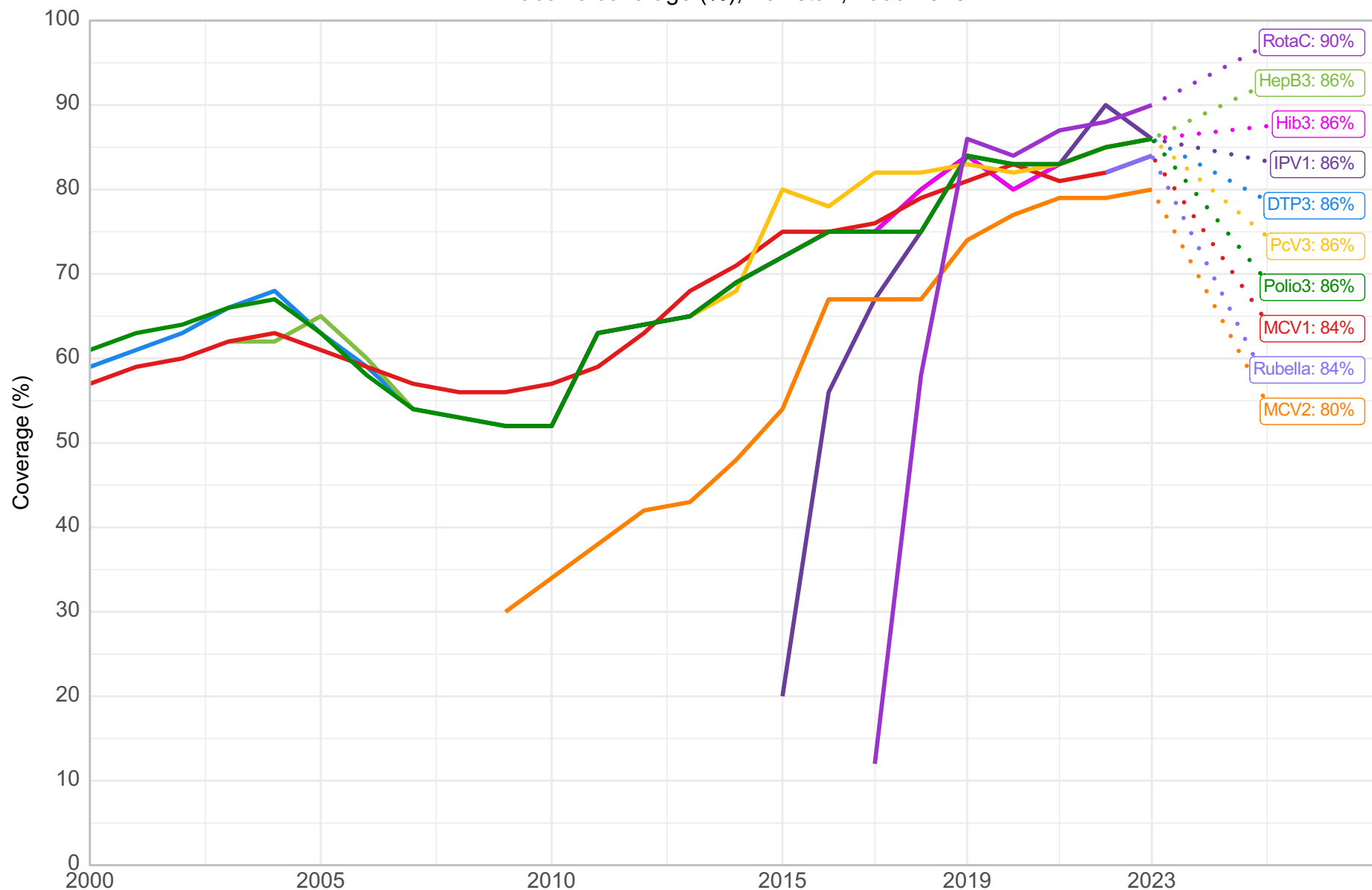
In 2023, MCV2 had the lowest coverage (80%), followed by MCV1 (84%).

Compared to 2019, coverage of 9 vaccines increased (BCG, DTP1, DTP3, IPV1, MCV1, MCV2, PCV3, POL3 and ROTAC).

Compared to 2022, coverage of 8 vaccines increased (BCG, DTP1, DTP3, MCV1, MCV2, PCV3, POL3 and ROTAC) and one vaccine decreased (IPV1).

Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision
Note: Data labels are shown for 2000 (or first year of reporting), 2019 and 2023

Vaccine coverage (%), Pakistan, 2000-2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision.
Numbers in the data label bubbles refer to vaccine coverage in the latest year estimates are available.

This chart shows trends in coverage of 9 vaccines (complete series).

In 2023, MCV2 had the lowest coverage of all vaccines (80%), followed by MCV1 (84%).

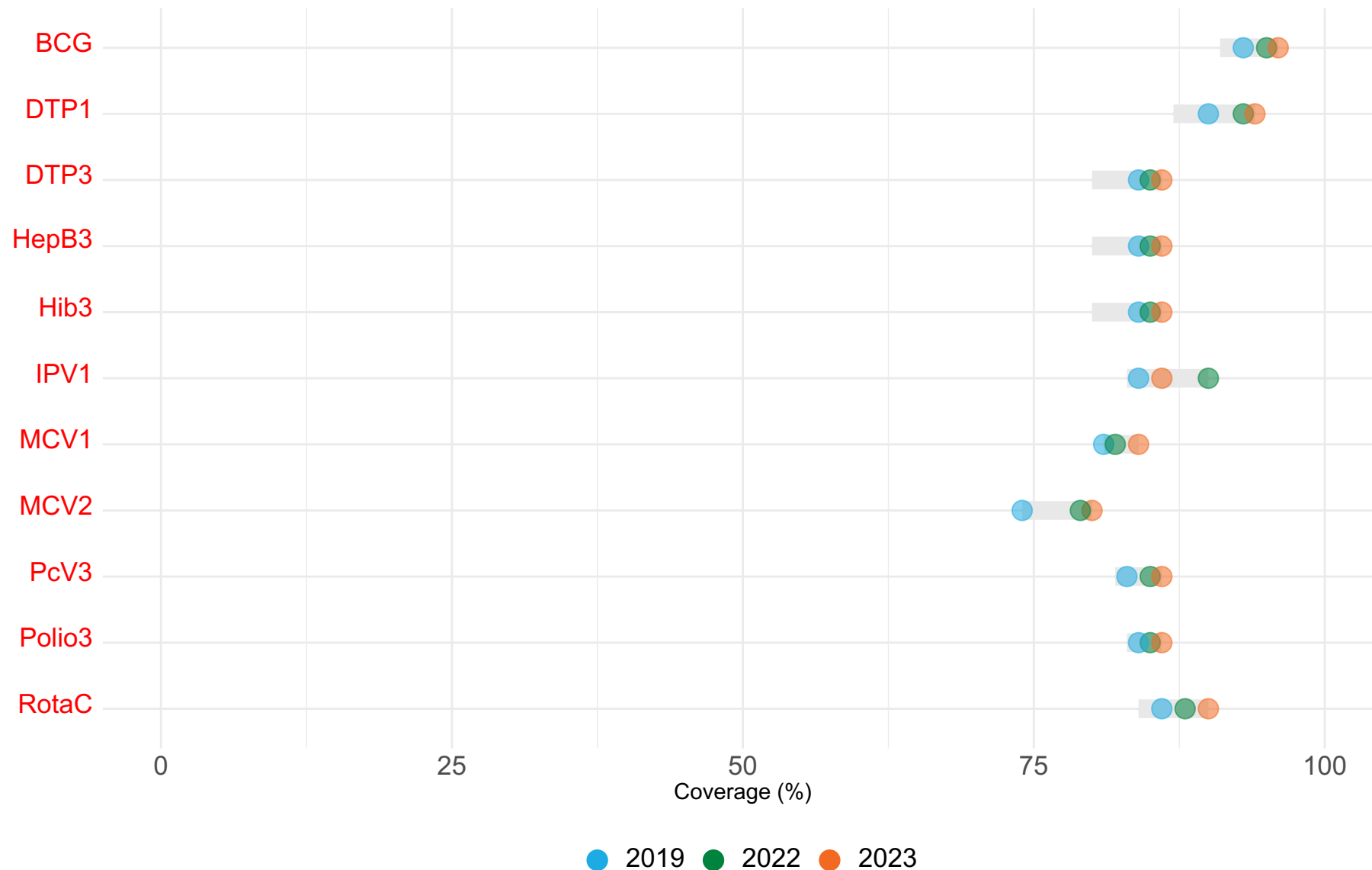
Coverage of 9 vaccines increased (DTP3, HepB3, Hib3, IPV1, MCV1, MCV2, PcV3, Polio3 and RotaC) compared to respective coverage in 2019.

Coverage of 9 vaccines increased (DTP3, HepB3, Hib3, MCV1, MCV2, PcV3, Polio3, RotaC and Rubella) and 1 vaccine decreased (IPV1) compared to respective coverage in 2022.



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Vaccine coverage, Pakistan, 2019-2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision.
 Note: The grey bar spans vaccine coverage across all years 2019-2023 and the dots represent coverage in specific years.
 Coverage is shown for vaccines with data all years 2019-2023.
 Vaccine names are coloured based on if coverage is lower (red), the same as (blue) or higher (green) than in 2019

This chart shows the range of coverage across all years 2019 to 2023 (grey bars), and coverage in specific years (dots), by vaccine. The chart can be used for assessing recovery to pre-pandemic levels.

DTP1 coverage increased between 2019 (90%) and 2022 (93%). DTP1 coverage increased in 2023 (94%) compared to 2022, and was higher than in 2019. In 2019-2023, DTP1 coverage was at it's lowest level in 2020 (87%).

In 2022, 0 vaccines had lower coverage than in 2019.

In 2023, 0 vaccines had lower coverage than in 2019.

In 2023, 2 vaccines had lower coverage than in 2022.

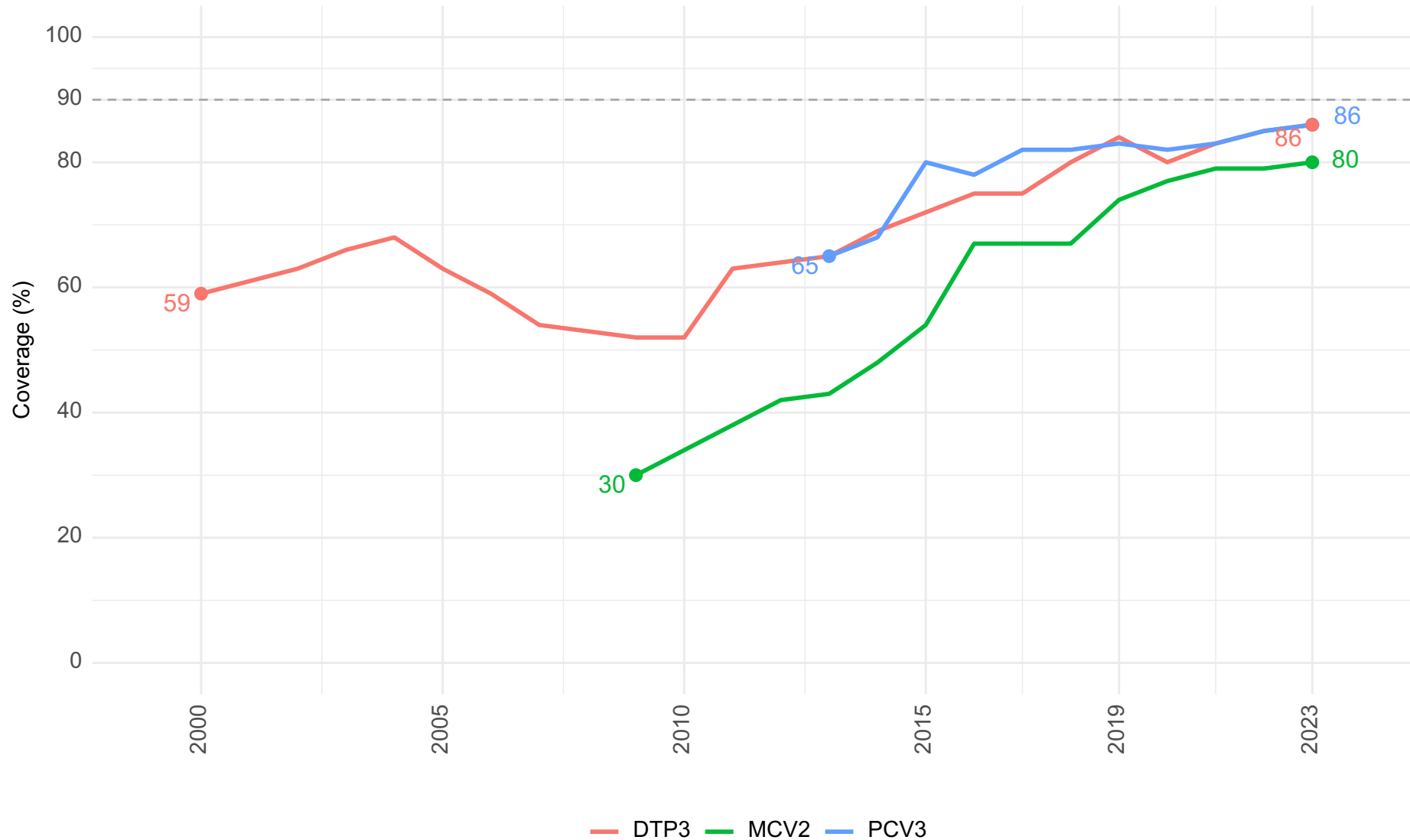
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SDG 3.b.1

SDG 3.b.1: Proportion of the target population covered by all vaccines included in their national programme, Pakistan, 2000-2023



Source: WHO/UNICEF Estimates of National Immunization Coverage, 2023 revision
Note: The four vaccination coverage indicators contribute to SDG indicator 3.b.1 are: DTP3, MCV2, PCV3 and HPVc
The Immunization Agenda 2030 (IA2030) global target is 90% coverage of all four antigens by 2030.
Data are missing for vaccines that have not been introduced by the country.

Four vaccination coverage indicators contribute to Sustainable Development Goal 3, indicator b.1: DTP3, PCV3, MCV2 and HPV.

The IA2030 global target is 90% coverage of all four antigens by 2030.

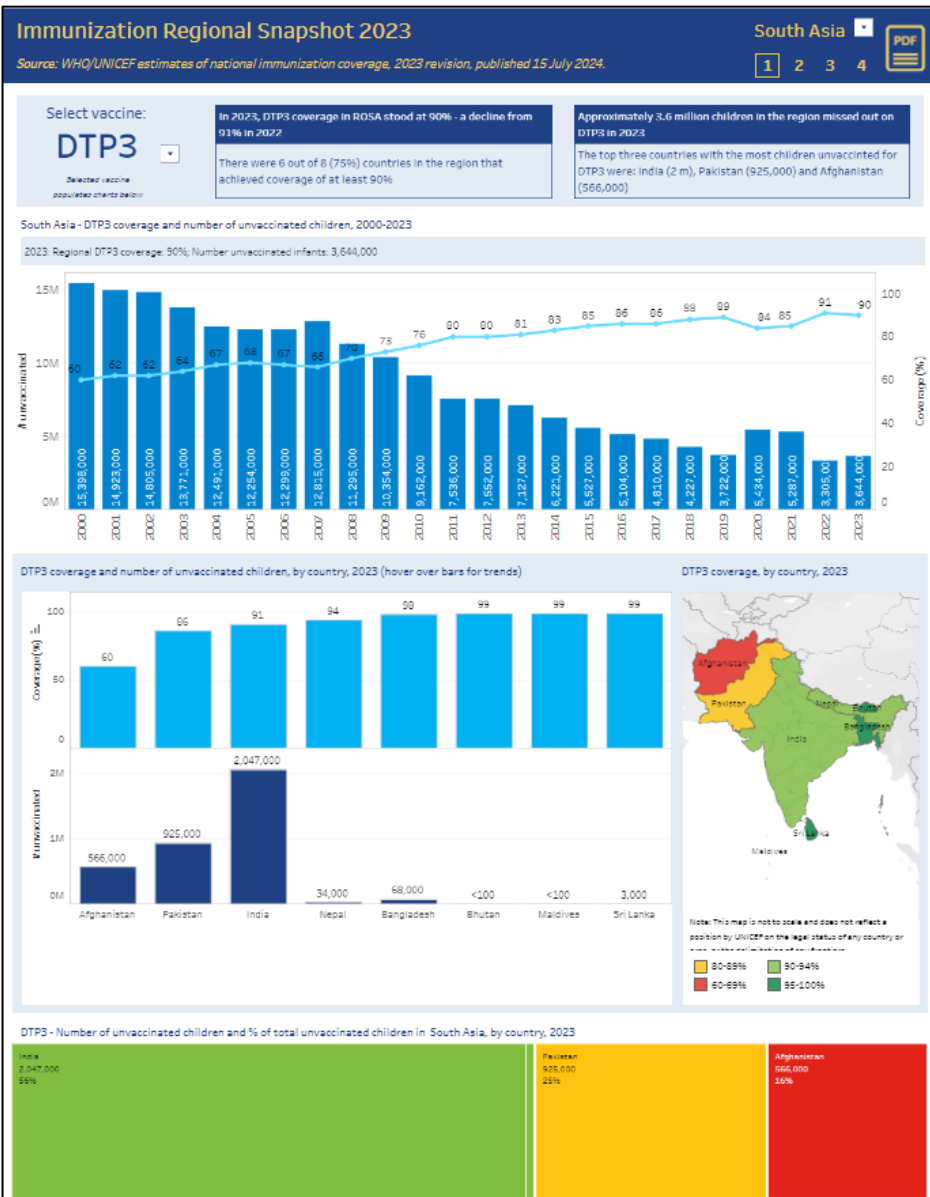
Pakistan has 3 out of the 4 SDG vaccines.

In 2023, Pakistan had achieved at least 90% coverage of none out of the 4 vaccines.



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Additional resources



Immunization Regional Snapshots

Interactive, multi-page profiles displaying national and regional WUENIC data by UNICEF region.

The dashboard includes:

- Key messages
- Trends in vaccine coverage and number of unvaccinated children,
- Comparisons between countries and regions,
- Maps
- Progress compared to 2019

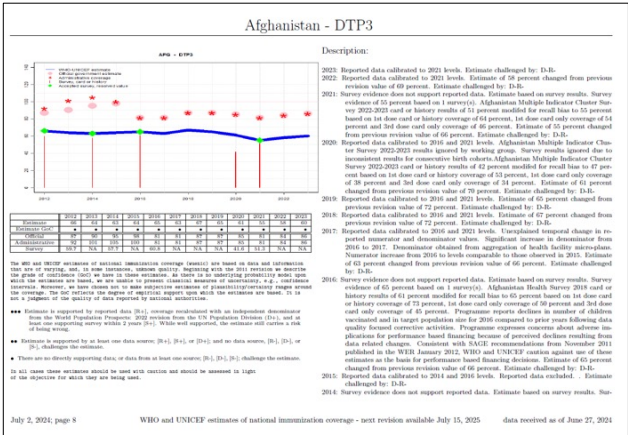
The profiles can be downloaded and are useful for users specifically interested in immunization data within UNICEF regions.

<https://data.unicef.org/resources/regional-immunization-snapshots/>

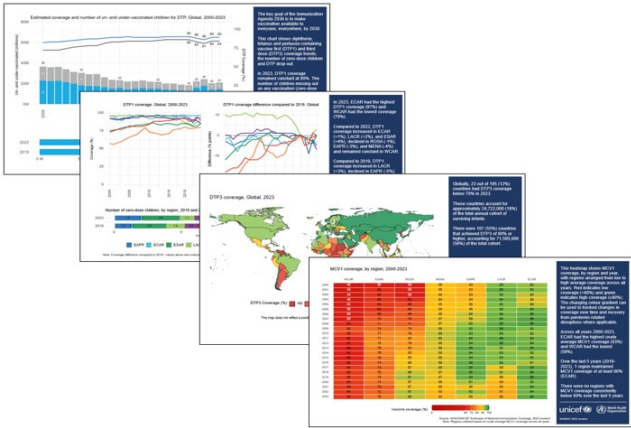
Interactive immunization regional snapshots: UNICEF regions



WUENIC country profiles



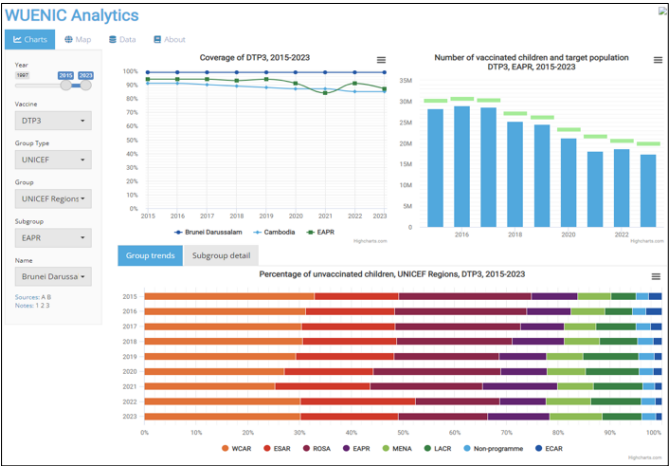
Regional slide decks: UNICEF, WHO, World Bank, Gavi and African Union regions



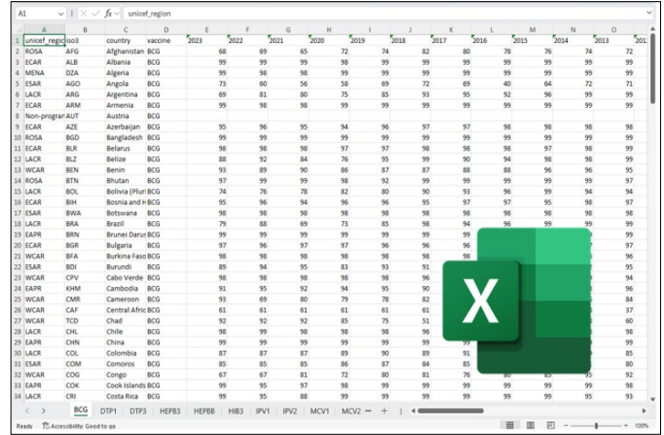
Interactive WUENIC country profiles



WUENIC Analytics Dashboard



Datasets: WUENIC, HPV, survey database



Additional resources

Additional immunization data resources can be found at: <https://data.unicef.org/resources/immunization/>

Interactive WUENIC country profiles can be found at: <https://worldhealthorg.shinyapps.io/wuenic-trends/>